

Airman Knowledge Testing Supplement for Instrument Rating

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Flight Standards Service

Preface

This Airman Knowledge Testing Supplement is designed by the Federal Aviation Administration (FAA) Flight Standards Service. It is intended for use by Airman Knowledge Testing (AKT) Organization Designation Authorization (ODA) Holders and other entities approved and/or authorized to administer airman knowledge tests on behalf of the FAA in the following knowledge areas:

Instrument Rating—Airplane (IRA)
Instrument Rating—Rotorcraft/Helicopter (IRH)
Instrument Rating—Powered Lift (IPL)
Instrument Flight Instructor—Powered Lift (IPI)
Instrument Rating—Foreign Pilot (IFP)
Instrument Flight Instructor—Airplane (FII)
Instrument Flight Instructor—Rotorcraft/Helicopter - (FIH)
Instrument Flight Instructor—Airplane (added rating) (AIF)
Instrument Flight Instructor—Rotorcraft/Helicopter (added rating) (HIF)
Ground Instructor—Instrument (IGI)

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APPENDIX 1

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GENERAL INFORMATION

ABBREVIATIONS

The following abbreviations/acronyms are those commonly used within this Directory. Other abbreviations/acronyms may be found in the Legend and are not duplicated below. The abbreviations presented are intended to represent grammatical variations of the basic form. (Example—"req" may mean "request", "requesting", "requested", or "requests").

Abbreviation	Description	Abbreviation	Description
A/G	air/ground	AM	Amplitude Modulation, midnight til noon
AAF	Army Air Field	AMC	Air Mobility Command
AAS	Airport Advisory Service	amdt	amendment
AB	Airbase	AMSL	Above Mean Sea Level
abm	abeam	ANGS	Air National Guard Station
ABn	Aerodrome Beacon	ant	antenna
abv	above	AOE	Airport/Aerodrome of Entry
ACC	Air Combat Command Area Control Center	AP	Area Planning
acft	aircraft	APAPI	Abbreviated Precision Approach Path Indicator
ACLS	Automatic Carrier Landing System	apch	approach
act	activity	apn	apron
ACWS	Aircraft Control and Warning Squadron	APP	Approach Control
ADA	Advisory Area	Apr	April
ADCC	Air Defense Control Center	aprx	approximate
ADCUS	Advise Customs	APU	Auxiliary Power Unit
addn	addition	apv, apvl	approve, approval
ADF	Automatic Direction Finder	ARB	Air Reserve Base
adj	adjacent	ARCAL (CANADA)	Aircraft Radio Control of Aerodrome Lighting
admin	administration	ARFF	Aircraft Rescue and Fire Fighting
ADR	Advisory Route	ARINC	Aeronautical Radio Inc
adv	advise	arng	arrange
advsy	advisory	arpt	airport
AEIS	Aeronautical Enroute Information Service	arr	arrive
AER	approach end runway	ARS	Air Reserve Station
AFA	Army Flight Activity	ARSA	Airport Radar Service Area
AFB	Air Force Base	ARSR	Air Route Surveillance Radar
afct	affect	ARTCC	Air Route Traffic Control Center
AFFF	Aqueous Film Forming Foam	AS	Air Station
AFHP	Air Force Heliport	ASAP	as soon as possible
AFIS	Aerodrome Flight Information Service	ASDA	Accelerate-Stop Distance Available
afld	airfield	ASDE	Airport Surface Detection
AFOD	Army Flight Operations Detachment	ASDE-X	Airport Surface Detection Equipment-Model X
AFR	Air Force Regulation	asgn	assign
AFRC	Armed Forces Reserve Center/Air Force Reserve Command	ASL	Above Sea Level
AFRS	American Forces Radio Stations	ASOS	Automated Surface Observing System
AFS	Air Force Station	ASR	Airport Surveillance Radar
AFSS	Automated Flight Service Station	ASSC	Airport Surface Surveillance Capability
AFTN	Aeronautical Fixed Telecommunication Network	ASU	Aircraft Starting Unit
AG	Agriculture	ATA	Actual Time of Arrival
A-G, A-GEAR	Arresting Gear	ATC	Air Traffic Control
agcy	Agency	ATCC	Air Traffic Control Center
AGL	above ground level	ATCT	Airport Traffic Control Tower
AHP	Army heliport	ATD	Actual Time of Departure Along Track Distance
AID	Airport Information Desk	ATIS	Automatic Terminal Information Service
AIS	Aeronautical Information Services	ATS	Air Traffic Service
AL	Approach and Landing Chart	attn	attention
ALF	Auxiliary Landing Field	Aug	August
ALS	Approach Light System	auth	authority
ALSF-1	High Intensity ALS Category I configuration with sequenced Flashers (code)	auto	automatic
ALSF-2	High Intensity ALS Category II configuration with sequenced Flashers (code)	AUW	All Up Weight (gross weight)
alt	altitude	aux	auxiliary
altn	alternate	AVASI	abbreviated VASI
		avbl	available
		AvGas	Aviation gasoline
		avn	aviation
		AvOil	aviation oil

LEGEND 1.—Abbreviations.

GENERAL INFORMATION

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Abbreviation	Description	Abbreviation	Description
AWOS	Automatic Weather Observing System	cdr	corridor
AWSS	Automated Weather Sensor System	cross	cross
awt	await	CRP	Compulsory Reporting Point
awy	airway	crs	course
az	azimuth	CS	call sign
		CSTMS	Customs
BA	braking action	CTA	Control Area
BASH	Bird Aircraft Strike Hazard	CTAF	Common Traffic Advisory Frequency
BC	back course	ctc	contact
bcn	beacon	ctl	control
bcst	broadcast	CTLZ	Control Zone
bdry	boundary	CVFR	Controlled Visual Flight Rules Areas
bldg	building	CW	Clockwise, Continuous Wave, Carrier Wave
blkd	blocked		
blo, blw	below	dalgt	daylight
BOQ	Bachelor Officers Quarters	D-ATIS	Digital Automatic Terminal Information Service
brg	bearing	daylt	daylight
btn	between	db	decibel
bus	business	DCL	Departure Clearance
byd	beyond	Dec	December
		decom	decommission
C	Commercial Circuit (Telephone)	deg	degree
CAC	Centralized Approach Control	del	delivery
cap	capacity	dep	depart
cat	category	DEP	Departure Control
CAT	Clear Air Turbulence	destn	destination
CCW or cntclkws	counterclockwise	det	detachment
ceil	ceiling	DF	Direction Finder
CERAP	Center Radar Approach Control	DH	Decision Height
CG	Coast Guard	DIAP	DoD Instrument Approach Procedure
CGAF	Coast Guard Air Facility	direc	directional
CGAS	Coast Guard Air Station	disem	disseminate
CH, chan	channel	displ	displace
CHAPI	Chase Helicopter Approach Path Indicator	dist	district, distance
chg	change	div	division
cht	chart	DL	Direct Line to FSS
cir	circle, circling	dl	delete
CIV, civ	Civil, civil, civilian	dly	daily
ck	check	DME	Distance Measuring Equipment (UHF standard, TACAN compatible)
CL	Centerline Lighting System	DNVT	Digital Non-Secure Voice Telephone
cl	class	DoD	Department of Defense
clnc	clearance	drct	direct
clsd	closed	DSN	Defense Switching Network (Telephone)
CNATRA	Chief of Naval Air Training	DSN	Defense Switching Network
cnl	cancel	dsplcd	displaced
cntr	center	DT	Daylight Savings Time
cntrln	centerline	dur	during
Co	Company, County	durn	duration
CO	Commanding Officer	DV	Distinguished Visitor
com	communication		
comd	command	E	East
Comdr	Commander	ea	each
coml	commercial	EAT	Expected Approach Time
compul	compulsory	ECN	Enroute Change Notice
comsn	commission	EFAS	Enroute Flight Advisory Service
conc	concrete	eff	effective, effect
cond	condition	E-HA	Enroute High Altitude
const	construction	E-LA	Enroute Low Altitude
cont	continue	elev	elevation
CONUS	Continental United States	ELT	Emergency Locator Transmitter
convl	conventional		
coord	coordinate	EMAS	Engineered Material Arresting System
copter	helicopter	emerg	emergency
corr	correct	eng	engine
CPDLC	Controller Pilot Data Link Communication		

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GENERAL INFORMATION

Abbreviation	Description	Abbreviation	Description
EOR	End of Runway	govt	government
eqpt	equipment	GP	Glide Path
ERDA	Energy Research and Development Administration	Gp	Group
E-S	Enroute Supplement	GPI	Ground Point of Intercept
est	estimate	grad	gradient
estab	establish	grd	guard
ETA	Estimated Time of Arrival	GS	glide slope
ETD	Estimated Time of Departure	GWT	gross weight
ETE	Estimated Time Enroute	H	Enroute High Altitude Chart (followed by identification)
ETS	European Telephone System	H+	Hours or hours plus...minutes past the hour
EUR	European (ICAO Region)	H24	continuous operation
ev	every	HAA	Height Above Airport/Aerodrome
evac	evacuate	HAL	Height Above Landing Area
exc	except	HAR	Height Above Runway
excl	exclude	HAT	Height Above Touchdown
exer	exercise	haz	hazard
exm	exempt	hdg	heading
exp	expect	HDTA	High Density Traffic Airport/Aerodrome
extd	extend	HF	High Frequency (3000 to 30,000 KHz)
extn	extension	hgr	hangar
extv	extensive	hgt	height
F/W	Fixed Wing	hi	high
FAA	Federal Aviation Administration	HIRL	High Intensity Runway Lights
fac	facility	HIWAS	Hazardous Inflight Weather Advisory Service
FAWS	Flight Advisory Weather Service	HO	Service available to meet operational requirements
fax	facsimile	hol	holiday
FBO	Fixed Base Operator	HOLF	Helicopter Outlying Field
FCC	Flight Control Center	hosp	hospital
FCG	Foreign Clearance Guide	HQ	Headquarters
FCLP	field carrier landing practice	hr	hour
fcst	forecast	HS	Service available during hours of scheduled operations
Feb	February	hsg	housing
FIC	Flight Information Center	hvy	heavy
FIH	Flight Information Handbook	HW	Heavy Weight
FIR	Flight Information Region	hwy	highway
FIS	Flight Information Service	HX	station having no specific working hours
FL	flight level	Hz	Hertz (cycles per second)
fld	field	I	Island
flg	flashing	IAP	Instrument Approach Procedure
FLIP	Flight Information Publication	IAS	Indicated Air Speed
flt	flight	IAW	in accordance with
flw	follow	ICAO	International Civil Aviation Organization
FM	Fan Marker, Frequency Modulation	ident	identification
FOC	Flight Operations Center	IFF	Identification, Friend or Foe
FOD	Foreign Object Damage	IFR	Instrument Flight Rules
fone	telephone	IFR-S	FLIP IFR Supplement
FPL	Flight Plan	IFSS	International Flight Service Station
fpm	feet per minute	ILS	Instrument Landing System
fr	from	IM	Inner Marker
freq	frequency, frequent	IMC	Instrument Meteorological Conditions
Fri	Friday	IMG	Immigration
frng	firing	immed	immediate
FSC	Flight Service Center	inbd	inbound
FSS	Flight Service Station	Inc.	Incorporated
ft	foot	incl	include
ftr	fighter	incr	increase
GA	Glide Angle	indef	indefinite
gal	gallon	info	information
GAT	General Air Traffic (Europe-Asia)	inop	inoperative
GCA	Ground Control Approach	inst	instrument
GCO	Ground Communication Outlet		
glidr	glider		
GND	Ground Control		
gnd	ground		

LEGEND 1B.—Abbreviations.

GENERAL INFORMATION

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Abbreviation Description

instl.....	install
instr.....	instruction
int.....	intersection
intcntl.....	intercontinental
intcp.....	intercept
intl.....	international
intmt.....	intermittent
ints.....	intense, intensity
invof.....	in the vicinity of
irreg.....	Irregularly
Jan.....	January
JASU.....	Jet Aircraft Starting Unit
JATO.....	Jet Assisted Take-Off
JOAP.....	Joint Oil Analysis Program
JOSAC.....	Joint Operational Support Airlift Center
JRB.....	Joint Reserve Base
Jul.....	July
Jun.....	June
K or Kt.....	Knots
kHz.....	kilohertz
KIAS.....	Knots Indicated Airspeed
KLIZ.....	Korea Limited Identification Zone
km.....	Kilometer
kw.....	kilowatt
L.....	Compass locator (Component of ILS system) under 25 Watts, 15 NM, Enroute Low Altitude Chart (followed by identification)
L.....	Local Time
LAA.....	Local Airport Advisory
LAHSO.....	Land and Hold-Short Operations
L-AOE.....	Limited Airport of Entry
LAWRS.....	Limited Aviation Weather Reporting Station
lb, lbs.....	pound (weight)
LC.....	local call
lcl.....	local
LCP.....	French Peripheral Classification Line
lctd.....	located
lctn.....	location
lctr.....	locator
LCVASI.....	Low Cost Visual Approach Slope Indicator
lczr.....	localizer
LD.....	long distance
LDA.....	Landing Distance Available
ldg.....	landing
LDIN.....	Lead-in Lights
LDOCF.....	Long Distance Operations Control Facility
len.....	length
lgt, lgtd, lgts.....	light, lighted, lights
LIRL.....	Low Intensity Runway Lights
LLWAS.....	Low-Level Wind Shear Alert System
LLZ.....	Localizer (Instrument Approach Procedures Identification only)
LMM.....	Compass locator at Middle Marker ILS
lo.....	low
LoALT or LA.....	Low Altitude
LOC.....	Localizer
LOM.....	Compass locator at Outer Marker ILS
LR.....	Long Range, Lead Radial
LRA.....	Landing Rights Airport
LRRS.....	Long Range RADAR Station

Abbreviation Description

LSB.....	lower side band
ltd.....	limited
M.....	meters, magnetic (after a bearing), Military Circuit (Telephone)
MACC.....	Military Area Control Center
mag.....	magnetic
maint.....	maintain, maintenance
maj.....	major
MALS.....	Medium Intensity Approach Lighting System
MALSF.....	MALS with Sequenced Flashers
MALSR.....	MALS with Runway Alignment Indicator Lights
Mar.....	March
MARA.....	Military Activity Restricted Area
MATO.....	Military Air Traffic Operations
MATZ.....	Military Aerodrome Traffic Zone
max.....	maximum
mb.....	millibars
MCAC.....	Military Common Area Control
MCAF.....	Marine Corps Air Facility
MCALF.....	Marine Corps Auxiliary Landing Field
MCAS.....	Marine Corps Air Station
MCB.....	Marine Corps Base
MCC.....	Military Climb Corridor
MCOLF.....	Marine Corps Outlying Field
MDA.....	Minimum Descent Altitude
MEA.....	Minimum Enroute Altitude
med.....	medium
MEHT.....	Minimum Eye Height over Threshold
mem.....	memorial
MET.....	Meteorological, Meteorology
METAR.....	Aviation Routine Weather Report (in international MET figure code)
METRO.....	Pilot-to-Metro voice cell
MF.....	Medium Frequency (300 to 3000 KHz), Mandatory Frequency (Canada)
MFA.....	Minimum Flight Altitude
mgmt.....	Management
mgr.....	manager
MHz.....	Megahertz
mi.....	mile
MID/ASIA.....	Middle East/Asia (ICAO Region)
MIJI.....	Meaconing, Intrusion, Jamming, and Interference
Mil, mil.....	military
min.....	minimum, minute
MIRL.....	Medium Intensity Runway Lights
misl.....	missile
mkr.....	marker (beacon)
MM.....	Middle Marker of ILS
mnt.....	monitor
MOA.....	Military Operations Area
MOCA.....	Minimum Obstruction Clearance Altitude
mod.....	modify
MOG.....	Maximum (aircraft) on the Ground
Mon.....	Monday
MP.....	Maintenance Period
MR.....	Medium Range
MRA.....	Minimum Reception Altitude
mrk.....	mark, marker
MSAW.....	minimum safe altitude warning
msg.....	message
MSL.....	Mean Sea Level
msn.....	Mission

LEGEND 1C.—Abbreviations.

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GENERAL INFORMATION

Abbreviation	Description	Abbreviation	Description
mt	mount, mountain	ODALS	Omnidirectional Approach Lighting System
MTAF	Mandatory Traffic Advisory Frequency	ODO	Operations Duty Officer
MTCA	Military Terminal Control Area	offl	official
mtlly	monthly	OIC	Officer In Charge
MUAC	Military Upper Area Control	OLF	Outlying Field
muni	municipal	OLS	Optical Landing System
MWARA	Major World Air Route Area	OM	Outer Marker, ILS
N	North	opr	operate, operator, operational
N/A	not applicable	OPS, ops	operations
NA	not authorized (For Instrument Approach Procedure take-off and alternate MINIMA only)	orig	original
NAAS	Naval Auxiliary Air Station	OROCA	Off Route Obstruction Clearance Altitude
NADC	Naval Air Development Center	ORTCA	Off Route Terrain Clearance Altitude
NADEP	Naval Air Depot	OT	other times
NAEC	Naval Air Engineering Center	OTS	out of service
NAES	Naval Air Engineering Station	outbd	outbound
NAF	Naval Air Facility	ovft	overflight
NALCO	Naval Air Logistics Control Office	ovrn	overrun
NALF	Naval Auxiliary Landing Field	OX	oxygen
NALO	Navy Air Logistics Office	P/L	plain language
NAS	Naval Air Station	PAC	Pacific (ICAO Region)
NAT	North Atlantic (ICAO Region)	PAEW	personnel and equipment working
natl	national	PALS	Precision Approach and Landing System (NAVY)
nav	navigation	PAPI	Precision Approach Path Indicator
navaid	navigation aid	PAR	Precision Approach Radar
NAVMTO	Navy Material Transportation Office	para	paragraph
NAWC	Naval Air Warfare Center	parl	parallel
NAWS	Naval Air Weapons Station	pat	pattern
NCRP	Non-Compulsory Reporting Point	PAX	Passenger
NDB	Non-Directional Radio Beacon	PCL	pilot controlled lighting
NE	Northeast	pent	penetrate
nec	necessary	perm	permanent
NEW	Net Explosives Weight	perms	permission
ngt	night	pers	personnel
NM	nautical miles	PFC	Porous Friction Courses
nml	normal	PJE	Parachuting Activities/Exercises
NMR	nautical mile radius	p-line	power line
No or Nr	number	PM	Post meridian, noon til midnight
NOLF	Naval Outlying Field	PMRF	Pacific Missile Range Facility
NORDO	Lost communications or no radio installed/available in aircraft	PMSV	Pilot-to-Metro Service
NOTAM	Notice to Airmen	PN	prior notice
Nov	November	POB	persons on board
npi	non precision instrument	POL	Petrol, Oils and Lubricants
Nr or No	number	posn	position
NS	Naval Station	PPR	prior permission required
NS ABTMT	Noise Abatement	prcht	parachute
NSA	Naval Support Activity	pref	prefer
NSF	Naval Support Facility	prev	previous
NSTD, nstd	nonstandard	prim	primary
ntc	notice	prk	park
NVD	Night Vision Devices	PRM	Precision Runway Monitor
NVG	Night Vision Goggles	pro	procedure
NW	Northwest	proh	prohibited
NWC	Naval Weapons Center	pt	point
O/A	On or about	PTD	Pilot to Dispatcher
O/S	out of service	pub	publication
O/R	On Request	publ	publish
OAT	Operational Air Traffic	PVASI	Pulsating Visual Approach Slope Indicator
obsn	observation	pvt	private
obst	obstruction	pwr	power
OCA	Oceanic Control Area	QFE	Altimeter Setting above station
ocnl	occasional		
Oct	October		

LEGEND 1D.—Abbreviations.

GENERAL INFORMATION

9

Abbreviation	Description	Abbreviation	Description
QNE	Altimeter Setting of 29.92 inches which provides height above standard datum plane	SAR	Search and Rescue
QNH	Altimeter Setting which provides height above mean sea level	Sat	Saturday
qtrs	quarters	SAVASI	Simplified Abbreviated Visual Approach Slope Indicator
quad	quadrant	SAWRS	Supplement Aviation Weather Reporting Station
R/T	Radiotelephony	sby	standby
R/W	Rotary/Wing	Sched	scheduled services
RACON	Radar Beacon	sctr	sector
rad	radius, radial	SDF	Simplified Directional Facility
RAIL	Runway Alignment Indicator Lights	SE	Southeast
RAMCC	Regional Air Movement Control Center	sec	second, section
R-AOE	Regular Airport of Entry	secd	secondary
RAPCON	Radar Approach Control (USAF)	SELCAL	Selective Calling System
RATCF	Radar Air Traffic Control Facility (Navy)	SELF	Strategic Expeditionary Landing Field
RCAG	Remote Center Air to Ground Facility	SEng	Single Engine
RCAGL	Remote Center Air to Ground Facility Long Range	Sep	September
RCL	runway centerline	SFA	Single Frequency Approach
RCLS	Runway Centerline Light System	sfc	surface
RCO	Remote Communications Outlet	SFL	Sequence Flashing Lights
rcpt	reception	SFRA	Special Flight Rules Area
RCR	Runway Condition Reading	SID	Standard Instrument Departure
rcv	receive	SIDA	Secure Identification Display Area
rcvr	receiver	SIF	Selective Identification Feature
rdo	radio	sked	schedule
reconst	reconstruct	SM	statute miles
reful	refueling	SOAP	Spectrometric Oil Analysis Program
reg	regulation, regular	SOF	Supervisor of Flying
REIL	Runway End Identifier Lights	SPB	Seaplane Base
rel	reliable	SR	sunrise
relctd	relocated	SRE	Surveillance Radar Element of GCA (Instrument Approach Procedures Identification only)
REP	Reporting Point	SS	sunset
req	request	SSALS/R	Simplified Short Approach Lighting System/with RAIL
RETIL	Rapid Exit Taxiway Indicator Light	SSB	Single Sideband
Rgn	Region	SSR	Secondary Surveillance Radar
Rgnl	Regional	STA	Straight-in Approach
rgt	right	std	standard
rgt tfc	right traffic	stn	station
rlgd	realigned	stor	storage
RLLS	Runway Lead-in Light System	str-in	Straight-in
rmk	remark	stu	student
rng	range, radio range	subj	subject
RNP	Required Navigation Performance	sum	summer
RON	Remain Overnight	Sun	Sunday
Rot Lt or Bcn	Rotating Light or Beacon	sur	surround
RPI	Runway Point of Intercept	survl	survival, surveillance
rpt	report	suspd	suspended
rqr	require	svc	service
RR	Railroad	svcg	servicing
RRP	Runway Reference Point	SW	Southwest
RSC	Runway Surface Condition	sys	system
RSDU	Radar Storm Detection Unit	TA	Transition Altitude
RSE	Runway Starter Extension/Starter Strip	TAC	Tactical Air Command
RSRS	Reduced Same Runway Separation	TAF	Aerodrome (terminal or alternate) forecast in abbreviated form
rstd	restricted	TALCE	Tanker Aircraft Control Element
rte	route	TCA	Terminal Control Area
ruf	rough	TCH	Threshold Crossing Height
RVR	Runway Visual Range	TCTA	Transcontinental Control Area
RVSM	Reduced Vertical Separation Minima	TD	Touchdown
rwy	runway	TDWR	Terminal Doppler Weather Radar
S	South	TDZ	Touchdown Zone
S/D	Seadrome	TDZL	Touchdown Zone Lights
SALS	Short Approach Lighting System		

LEGEND 1E.—Abbreviations.

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GENERAL INFORMATION

Abbreviation Description

tfc.....	traffic
thld.....	threshold
thou.....	thousand
thru.....	through
Thu.....	Thursday
til.....	until
tkf, tkof.....	take-off
TLv.....	Transition Level
tmpry.....	temporary
TODA.....	Take-Off Distance Available
TORA.....	Take-Off Run Available
TP.....	Tire Pressure
TPA.....	Traffic Pattern Altitude
TRACON.....	Terminal Radar Approach Control (FAA)
tran.....	transient
trans.....	transmit
trml.....	terminal
trng.....	training
trns.....	transition
TRSA.....	Terminal Radar Service Area
Tue.....	Tuesday
TV.....	Television
TWEB.....	Transcribed Weather Broadcast
twr.....	tower
twy.....	taxiway
UACC.....	Upper Area Control Center (used outside US)
UAS.....	Unmanned Aerial Systems
UC.....	Under Construction
UCN.....	Urgent Change Notice
UDA.....	Upper Advisory Area
UDF.....	Ultra High Frequency Direction Finder
UFN.....	until further notice
UHF.....	Ultra High Frequency (300 to 3000 MHz)
UIR.....	Upper Flight Information Region
unauthd.....	unauthorized
unavbl.....	unavailable
unctl.....	uncontrolled
unk.....	unknown
unlgtd.....	unlighted
unltd.....	unlimited
unmrk.....	unmarked
unmto.....	unmonitored
unrel.....	unreliable
unrstd.....	unrestricted
unsatfy.....	unsatisfactory
unsked.....	unscheduled
unsvc.....	unserviceable
unuse, unusbl.....	unusable
USA.....	United States Army
USAF.....	United States Air Force
USB.....	Upper Side Band
USCG.....	United States Coast Guard
USMC.....	United States Marine Corps
USN.....	United States Navy
UTA.....	Upper Control Area
UTC.....	Coordinated Universal Time
V.....	Defense Switching Network (telephone, formerly AUTOVON)
VSTOL.....	Vertical and Short Take-off and Landing aircraft
VAL.....	Visiting Aircraft Line
var.....	variation (magnetic variation)
VASI.....	Visual Approach Slope Indicator

Abbreviation..... Description

vcnty.....	vicinity
VDF.....	Very High Frequency Direction Finder
veh.....	vehicle
vert.....	vertical
VFR.....	Visual Flight Rules
VFR-S.....	FLIP VFR Supplement
VHF.....	Very High Frequency (30 to 300 MHz)
VIP.....	Very Important Person
vis.....	visibility
VMC.....	Visual Meteorological Conditions
VOIP.....	Voice Over Internet Protocol
VOLMET.....	Meteorological Information for Aircraft in Flight
VOT.....	VOR Receiver Testing Facility
W.....	Warning Area (followed by identification), Watts, West, White
WCH.....	Wheel Crossing Height
Wed.....	Wednesday
Wg.....	Wing
WIE.....	with immediate effect
win.....	winter
WIP.....	work in progress
WSO.....	Weather Service Office
WSFO.....	Weather Service Forecast Office
wk.....	week
wkd.....	weekday
wkly.....	weekly
wng.....	warning
wo.....	without
WSP.....	Weather System Processor
wt.....	weight
wx.....	weather
yd.....	yard
yr.....	year
Z.....	Greenwich Mean Time (time groups only)

LEGEND 1F.—Abbreviations.

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AIRPORT/FACILITY DIRECTORY LEGEND

SAMPLE

1
CITY NAME

2
AIRPORT NAME (ALTERNATE NAME) (LTS)(KLTS) CIV/MIL 3 N UTC-6(-5DT) N34°41.93' W99°20.20'

3
200 B TPA-1000(800) AOE LRA Class IV, ARFF Index A NOTAM FILE ORL Not insp.

4
JACKSONVILLE COPTER

5
H-4G, L-19C

6
IAP, DIAP, AD

7
RWY 18-36: H12004X200 (ASPH-CONC-GRVD)

8
S-90, D-160, 2D-300 PCN 80 R/B/W/T HIRL CL

9
RWY 18: RLLS. MALSF. TDZL. REIL. PAPI(P2R)-GA 3.0° TCH 36'

10
RVR-TMR. Thld displd 300'. Trees. Rgt tfc. 0.3% up.

11
RWY 36: ALSF1. 0.4% down.

12
RWY 09-27: H6000X150 (ASPH) MIRL

13
RWY 173-353: H3515X150 (ASPH-PFC) AUW PCN 59 F/A/W/T

14
LAND AND HOLD-SHORT OPERATIONS

15
LDG RWY HOLD-SHORT POINT AVBL LDG DIST

16
RWY 18 09-27 6500

17
RWY 36 09-27 5400

18
RUNWAY DECLARED DISTANCE INFORMATION

19
RWY 18: TORA-12004 TODA-12004 ASDA-11704 LDA-11504

20
RWY 36: TORA-12004 TODA-12004 ASDA-12004 LDA-11704

21
ARRESTING GEAR/SYSTEM

22
RWY 18 HOOK E5 (65' OVRN) BAK-14 BAK-12B (1650')

23
BAK-14 BAK-12B (1087') HOOK E5 (74' OVRN) RWY 36

24
SERVICE: S4 FUEL 100LL, JET A OX 1, 3 LGT ACTIVATE MALSR Rwy

25
29, REIL Rwy 11, VASI Rwy 11, HIRL Rwy 11-29, PAPI Rwy 17

26
and Rwy 35, MIRL Rwy 17-35-CTAF. MILITARY- A-GEAR E-5

27
connected on dep end, disconnected on apch end.

28
JASU 3(AM32A-60) 2(A/M32A-86) FUEL J8(Mil)(NC-100, A)

29
FLUID W SP PRESAIR LOX OIL O-128 MAINT S1 Mon-Fri 1000-2200Z†

30
TRAN ALERT Avbl 1300-0200Z† svc limited weekends.

31
AIRPORT REMARKS: Special Air Traffic Rules-Part 93, see Regulatory Notices. Attended 1200-0300Z†. Parachute Jumping.

Deer invof arpt. Heavy jumbo jet training surface to 9000'. Twy A clsd indef. Flight Notification Service (ADCUS) avbl.

MILITARY REMARKS: ANG PPR/Official Business Only. Base OPS DSN 638-4390, C503-335-4222. Ctc Base OPS 15 minutes prior to ldg and after dep. Limited tran parking.

AIRPORT MANAGER: (580) 481-5739

WEATHER DATA SOURCES: AWOS-1 120.3 (202) 426-8000. LAWRS.

COMMUNICATIONS: SFA CTAF 122.8 UNICOM 122.95 ATIS 127.25 273.5 (202) 426-8003 PTD 372.2

NAME FSS (ORL) on arpt. 123.65 122.65 122.2

NAME RCO 112.2T 112.1R (NAME RADIO)

NAME APP/DEP CON 128.35 257.725 (1200-0400Z†)

TOWER 119.65 255.6 (1200-0400Z†) GND CON 121.7 GCO 135.075 (ORLANDO CLNC) CLNC DEL 125.55

CPDLC D-HZWXR, D-TAXI, DCL (LOGON KMEN)

NAME COMD POST (GERONIMO) 311.0 321.4 6761 PMSV METRO 239.8 NAME OPS 257.5

AIRSPACE: CLASS B See VFR Terminal Area Chart.

VOR TEST FACILITY (VOT): 116.7

RADIO AIDS TO NAVIGATION: NOTAM FILE ORL. VHF/DF ctc FSS.

(H) VORTAC 112.2 MCO Chan 59 N28°32.55' W81°20.12' at fld. 1110/8E.

(H) TACAN Chan 29 CBU (109.2) N28°32.65' W81°21.12' at fld. 1115/8E.

HERNY NDB (LOM) 221 OR N28°37.40' W81°21.05' 177° 5.4 NM to fld.

ILS/DME 108.5 I-ORL Chan 22 Rwy 18. Class IIE. LOM HERNY NDB.

ASR/PAR (1200-0400Z†)

COMM/NAV/WEATHER REMARKS: Emerg frequency 121.5 not avbl at twr.

HELIPAD H1: H100X75 (ASPH)

HELIPAD H2: H60X60 (ASPH)

HELIPORT REMARKS: Helipad H1 lctd on general aviation side and H2 lctd on air carrier side of arpt.

187 TPA 1000(813)

WATERWAY 15-33: 5000X425 (WATER)

SEAPLANE REMARKS: Birds roosting and feeding areas along river banks. Seaplanes operating adjacent to SW side of arpt not visible from twr and are required to ctc twr.

All bearings and radials are magnetic unless otherwise specified. All mileages are nautical unless otherwise noted.
All times are Coordinated Universal Time (UTC) except as noted. All elevations are in feet above/below Mean Sea Level (MSL) unless otherwise noted.
The horizontal reference datum of this publication is North American Datum of 1983 (NAD83), which for charting purposes is considered equivalent to World Geodetic System 1984 (WGS 84).

LEGEND 2.—Airport/Facility Directory.

AIRPORT/FACILITY DIRECTORY LEGEND

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SKETCH LEGEND

RUNWAYS/LANDING AREAS

Hard Surfaced	
Metal Surface	
Sod, Gravel, etc.	
Light Plane,	
Ski Landing Area or Water	
Under Construction	
Closed	
Helicopter Landings Area	
Displaced Threshold	
Taxiway, Apron and Stopways ..	

MISCELLANEOUS BASE AND CULTURAL FEATURES

Buildings	
Power Lines	
Fence	
Towers	
Wind Turbine	
Tanks	
Oil Well	
Smoke Stack	
Obstruction	
Controlling Obstruction	
Trees	
Populated Places	
Cuts and Fills	
Cliffs and Depressions ..	
Ditch	
Hill	

RADIO AIDS TO NAVIGATION

VORTAC ...		VOR	
VOR/DME ..		NDB	
TACAN		NDB/DME	
DME			

MISCELLANEOUS AERONAUTICAL FEATURES

Airport Beacon	
Wind Cone	
Landing Tee	
Tetrahedron	
Control Tower	or TWR

When control tower and rotating beacon are co-located beacon symbol will be used and further identified as TWR.

APPROACH LIGHTING SYSTEMS

A dot "•" portrayed with approach lighting letter identifier indicates sequenced flashing lights (F) installed with the approach lighting system e.g., (A1) Negative symbology, e.g., (A1)

(V) indicates Pilot Controlled Lighting (PCL).

Runway Centerline Lighting	
(A) Approach Lighting System ALSF-2 ..	
(A1) Approach Lighting System ALSF-1 ..	
(A2) Short Approach Lighting System SALS/SALSF	
(A3) Simplified Short Approach Lighting System (SSALR) with RAIL	
(A4) Medium Intensity Approach Lighting System (MALS and MALSF)/(SSALS and SSALF)	
(A5) Medium Intensity Approach Lighting System (MALSR) and RAIL	
(V) Omnidirectional Approach Lighting System (ODALS)	
(D) Navy Parallel Row and Cross Bar ..	
(F) Air Force Overrun	
(V) Visual Approach Slope Indicator with Standard Threshold Clearance provided	
(V2) Pulsating Visual Approach Slope Indicator (PVASI)	
(V3) Visual Approach Slope Indicator with a threshold crossing height to accommodate long bodied or jumbo aircraft	
(V4) Tri-color Visual Approach Slope Indicator (TRCV)	
(V5) Approach Path Alignment Panel (APAP)	
(P) Precision Approach Path Indicator (PAPI)	

LEGEND 3.—Airport/Facility Directory.

AIRPORT/FACILITY DIRECTORY LEGEND**LEGEND**

This directory is a listing of data on record with the FAA on public-use airports, military airports and selected private-use airports specifically requested by the Department of Defense (DoD) for which a DoD Instrument Approach Procedure has been published in the U.S. Terminal Procedures Publication. Additionally this listing contains data for associated terminal control facilities, air route traffic control centers, and radio aids to navigation within the conterminous United States, Puerto Rico and the Virgin Islands. Civil airports and joint Civil/Military airports which are open to the public are listed alphabetically by state, associated city and airport name and cross-referenced by airport name. Military airports and private-use (limited civil access) joint Military/Civil airports are listed alphabetically by state and official airport name and cross-referenced by associated city name. Nav aids, flight service stations and remote communication outlets that are associated with an airport, but with a different name, are listed alphabetically under their own name, as well as under the airport with which they are associated.

The listing of an airport as open to the public in this directory merely indicates the airport operator's willingness to accommodate transient aircraft, and does not represent that the airport conforms with any Federal or local standards, or that it has been approved for use on the part of the general public. Military airports, private-use airports, and private-use (limited civil access) joint Military/Civil airports are open to civil pilots only in an emergency or with prior permission. See Special Notice Section, Civil Use of Military Fields.

The information on obstructions is taken from reports submitted to the FAA. Obstruction data has not been verified in all cases. Pilots are cautioned that objects not indicated in this tabulation (or on the airports sketches and/or charts) may exist which can create a hazard to flight operation. Detailed specifics concerning services and facilities tabulated within this directory are contained in the Aeronautical Information Manual, Basic Flight Information and ATC Procedures.

The legend items that follow explain in detail the contents of this Directory and are keyed to the circled numbers on the sample on the preceding pages.

① CITY/AIRPORT NAME

Civil and joint Civil/Military airports which are open to the public are listed alphabetically by state and associated city. Where the city name is different from the airport name the city name will appear on the line above the airport name. Airports with the same associated city name will be listed alphabetically by airport name and will be separated by a dashed rule line. A solid rule line will separate all others. FAA approved helipads and seaplane landing areas associated with a land airport will be separated by a dotted line. Military airports and private-use (limited civil access) joint Military/Civil airports are listed alphabetically by state and official airport name.

② ALTERNATE NAME

Alternate names, if any, will be shown in parentheses.

③ LOCATION IDENTIFIER

The location identifier is a three or four character FAA code followed by a four-character ICAO code, when assigned, to airports. If two different military codes are assigned, both codes will be shown with the primary operating agency's code listed first. These identifiers are used by ATC in lieu of the airport name in flight plans, flight strips and other written records and computer operations. Zeros will appear with a slash to differentiate them from the letter "O".

④ OPERATING AGENCY

Airports within this directory are classified into two categories, Military/Federal Government and Civil airports open to the general public, plus selected private-use airports. The operating agency is shown for military, private-use and joint use airports. The operating agency is shown by an abbreviation as listed below. When an organization is a tenant, the abbreviation is enclosed in parenthesis. No classification indicates the airport is open to the general public with no military tenant.

A	US Army	MC	Marine Corps
AFRC	Air Force Reserve Command	MIL/CIV	Joint Use Military/Civil Limited Civil Access
AF	US Air Force	N	Navy
ANG	Air National Guard	NAF	Naval Air Facility
AR	US Army Reserve	NAS	Naval Air Station
ARNG	US Army National Guard	NASA	National Air and Space Administration
CG	US Coast Guard	P	US Civil Airport Wherein Permit Covers Use by Transient Military Aircraft
CIV/MIL	Joint Use Civil/Military Open to the Public	PVT	Private Use Only (Closed to the Public)
DND	Department of National Defense Canada		

⑤ AIRPORT LOCATION

Airport location is expressed as distance and direction from the center of the associated city in nautical miles and cardinal points, e.g., 4 NE.

⑥ TIME CONVERSION

Hours of operation of all facilities are expressed in Coordinated Universal Time (UTC) and shown as "Z" time. The directory indicates the number of hours to be subtracted from UTC to obtain local standard time and local daylight saving time UTC-5(-4DT). The symbol ‡ indicates that during periods of Daylight Saving Time (DST) effective hours will be one hour earlier than shown. In those areas where daylight saving time is not observed the (-4DT) and ‡ will not be shown. Daylight saving time is in effect from 0200 local time the second Sunday in March to 0200 local time the first Sunday in November. Canada and all U.S. Conterminous States observe daylight saving time except Arizona and Puerto Rico, and the Virgin Islands. If the state observes daylight saving time and the operating times are other than daylight saving times, the operating hours will include the dates, times and no ‡ symbol will be shown, i.e., April 15-Aug 31 0630-1700Z, Sep 1-Apr 14 0600-1700Z.

LEGEND 4.—Airport/Facility Directory.

AIRPORT/FACILITY DIRECTORY LEGEND**15****⑦ GEOGRAPHIC POSITION OF AIRPORT—AIRPORT REFERENCE POINT (ARP)**

Positions are shown as hemisphere, degrees, minutes and hundredths of a minute and represent the approximate geometric center of all usable runway surfaces.

⑧ CHARTS

Charts refer to the Sectional Chart and Low and High Altitude Enroute Chart and panel on which the airport or facility is located. Helicopter Chart locations will be indicated as COPTER. IFR Gulf of Mexico West and IFR Gulf of Mexico Central will be depicted as GOMW and GOMC.

⑨ INSTRUMENT APPROACH PROCEDURES, AIRPORT DIAGRAMS

IAP indicates an airport for which a prescribed (Public Use) FAA Instrument Approach Procedure has been published. DIAP indicates an airport for which a prescribed DoD Instrument Approach Procedure has been published in the U.S. Terminal Procedures. See the Special Notice Section of this directory, Civil Use of Military Fields and the Aeronautical Information Manual 5–4–5 Instrument Approach Procedure Charts for additional information. AD indicates an airport for which an airport diagram has been published. Airport diagrams are located in the back of each Chart Supplement volume alphabetically by associated city and airport name.

⑩ AIRPORT SKETCH

The airport sketch, when provided, depicts the airport and related topographical information as seen from the air and should be used in conjunction with the text. It is intended as a guide for pilots in VFR conditions. Symbolology that is not self-explanatory will be reflected in the sketch legend. The airport sketch will be oriented with True North at the top. Airport sketches will be added incrementally.

⑪ ELEVATION

The highest point of an airport's usable runways measured in feet from mean sea level. When elevation is sea level it will be indicated as "00". When elevation is below sea level a minus "-" sign will precede the figure.

⑫ ROTATING LIGHT BEACON

B indicates rotating beacon is available. Rotating beacons operate sunset to sunrise unless otherwise indicated in the AIRPORT REMARKS or MILITARY REMARKS segment of the airport entry.

⑬ TRAFFIC PATTERN ALTITUDE

Traffic Pattern Altitude (TPA)—The first figure shown is TPA above mean sea level. The second figure in parentheses is TPA above airport elevation. Multiple TPA shall be shown as "TPA—See Remarks" and detailed information shall be shown in the Airport or Military Remarks Section. Traffic pattern data for USAF bases, USN facilities, and U.S. Army airports (including those on which ACC or U.S. Army is a tenant) that deviate from standard pattern altitudes shall be shown in Military Remarks.

⑭ AIRPORT OF ENTRY, LANDING RIGHTS, AND CUSTOMS USER FEE AIRPORTS

U.S. CUSTOMS USER FEE AIRPORT—Private Aircraft operators are frequently required to pay the costs associated with customs processing.

AOE—Airport of Entry. A customs Airport of Entry where permission from U.S. Customs is not required to land. However, at least one hour advance notice of arrival is required.

LRA—Landing Rights Airport. Application for permission to land must be submitted in advance to U.S. Customs. At least one hour advance notice of arrival is required.

NOTE: Advance notice of arrival at both an AOE and LRA airport may be included in the flight plan when filed in Canada or Mexico. Where Flight Notification Service (ADCUS) is available the airport remark will indicate this service. This notice will also be treated as an application for permission to land in the case of an LRA. Although advance notice of arrival may be relayed to Customs through Mexico, Canada, and U.S. Communications facilities by flight plan, the aircraft operator is solely responsible for ensuring that Customs receives the notification. (See Customs, Immigration and Naturalization, Public Health and Agriculture Department requirements in the International Flight Information Manual for further details.)

U.S. CUSTOMS AIR AND SEA PORTS, INSPECTORS AND AGENTS

Northeast Sector (New England and Atlantic States—ME to MD)	407-975-1740
Southeast Sector (Atlantic States—DC, WV, VA to FL)	407-975-1780
Central Sector (Interior of the US, including Gulf states—MS, AL, LA)	407-975-1760
Southwest East Sector (OK and eastern TX)	407-975-1840
Southwest West Sector (Western TX, NM and AZ)	407-975-1820
Pacific Sector (WA, OR, CA, HI and AK)	407-975-1800

⑮ CERTIFICATED AIRPORT (14 CFR PART 139)

Airports serving Department of Transportation certified carriers and certified under 14 CFR part 139 are indicated by the Class and the ARFF Index; e.g. Class I, ARFF Index A, which relates to the availability of crash, fire, rescue equipment. Class I airports can have an ARFF Index A through E, depending on the aircraft length and scheduled departures. Class II, III, and IV will always carry an Index A.

AIRPORT CLASSIFICATIONS

Type of Air Carrier Operation	Class I	Class II	Class III	Class IV
Scheduled Air Carrier Aircraft with 31 or more passenger seats	X			
Unscheduled Air Carrier Aircraft with 31 or more passengers seats	X	X		X
Scheduled Air Carrier Aircraft with 10 to 30 passenger seats	X	X	X	

LEGEND 5.—Airport/Facility Directory.

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AIRPORT/FACILITY DIRECTORY LEGEND

INDICES AND AIRCRAFT RESCUE AND FIRE FIGHTING EQUIPMENT REQUIREMENTS

Airport Index	Required No. Vehicles	Aircraft Length	Scheduled Departures	Agent + Water for Foam
A	1	<90'	≥1	500#DC or HALON 1211 or 450#DC + 100 gal H ₂ O
B	1 or 2	≥90', <126'	≥5	Index A + 1500 gal H ₂ O
		≥126', <159'	<5	
C	2 or 3	≥126', <159'	≥5	Index A + 3000 gal H ₂ O
		≥159', <200'	<5	
D	3	≥159', <200'		Index A + 4000 gal H ₂ O
		>200'	<5	
E	3	≥200'	≥5	Index A + 6000 gal H ₂ O

> Greater Than; < Less Than; ≥ Equal or Greater Than; ≤ Equal or Less Than; H₂O—Water; DC—Dry Chemical.

NOTE: The listing of ARFF index does not necessarily assure coverage for non-air carrier operations or at other than prescribed times for air carrier. ARFF Index Ltd.—indicates ARFF coverage may or may not be available, for information contact airport manager prior to flight.

16 NOTAM SERVICE

All public use landing areas are provided NOTAM service. A NOTAM FILE identifier is shown for individual landing areas, e.g., "NOTAM FILE BNA". See the AIM, Basic Flight Information and ATC Procedures for a detailed description of NOTAMs. Current NOTAMs are available from flight service stations at 1-800-WX-BRIEF (992-7433) or online through the FAA PilotWeb at <https://pilotweb.nas.faa.gov>. Military NOTAMs are available using the Defense Internet NOTAM Service (DINS) at <https://www.notams.faa.gov>. Pilots flying to or from airports not available through the FAA PilotWeb or DINS can obtain assistance from Flight Service.

17 FAA INSPECTION

All airports not inspected by FAA will be identified by the note: Not insp. This indicates that the airport information has been provided by the owner or operator of the field.

18 RUNWAY DATA

Runway information is shown on two lines. That information common to the entire runway is shown on the first line while information concerning the runway ends is shown on the second or following line. Runway direction, surface, length, width, weight bearing capacity, lighting, and slope, when available are shown for each runway. Multiple runways are shown with the longest runway first. Direction, length, width, and lighting are shown for sea-lanes. The full dimensions of helipads are shown, e.g., 50X150. Runway data that requires clarification will be placed in the remarks section.

RUNWAY DESIGNATION

Runways are normally numbered in relation to their magnetic orientation rounded off to the nearest 10 degrees. Parallel runways can be designated L (left)/R (right)/C (center). Runways may be designated as Ultralight or assault strips. Assault strips are shown by magnetic bearing.

RUNWAY DIMENSIONS

Runway length and width are shown in feet. Length shown is runway end to end including displaced thresholds, but excluding those areas designed as overruns.

RUNWAY SURFACE AND SURFACE TREATMENT

Runway lengths prefixed by the letter "H" indicate that the runways are hard surfaced (concrete, asphalt, or part asphalt-concrete). If the runway length is not prefixed, the surface is sod, clay, etc. The runway surface composition is indicated in parentheses after runway length as follows:

(AFSC)—Aggregate friction seal coat	(GRVL)—Gravel, or cinders	(SAND)—Sand
(AM2)—Temporary metal planks coated with nonskid material	(MATS)—Pierced steel planking, landing mats, membranes	(TURF)—Turf
(ASPH)—Asphalt	(PEM)—Part concrete, part asphalt	(TRTD)—Treated
(CONC)—Concrete	(PFC)—Porous friction courses	(WC)—Wire combed
(DIRT)—Dirt	(PSP)—Pierced steel plank	
(GRVD)—Grooved	(RFSC)—Rubberized friction seal coat	

LEGEND 6.—Airport/Facility Directory.

AIRPORT/FACILITY DIRECTORY LEGEND**17****RUNWAY WEIGHT BEARING CAPACITY**

Runway strength data shown in this publication is derived from available information and is a realistic estimate of capability at an average level of activity. It is not intended as a maximum allowable weight or as an operating limitation. Many airport pavements are capable of supporting limited operations with gross weights in excess of the published figures. Permissible operating weights, insofar as runway strengths are concerned, are a matter of agreement between the owner and user. When desiring to operate into any airport at weights in excess of those published in the publication, users should contact the airport management for permission. Runway strength figures are shown in thousand of pounds, with the last three figures being omitted. Add 000 to figure following S, D, 2S, 2T, AUW, SWL, etc., for gross weight capacity. A blank space following the letter designator is used to indicate the runway can sustain aircraft with this type landing gear, although definite runway weight bearing capacity figures are not available, e.g., S, D. Applicable codes for typical gear configurations with S=Single, D=Dual, T=Triple and Q=Quadruple:

CURRENT	NEW	NEW DESCRIPTION
S	S	Single wheel type landing gear (DC3), (C47), (F15), etc.
D	D	Dual wheel type landing gear (BE1900), (B737), (A319), etc.
T	D	Dual wheel type landing gear (P3, C9).
ST	2S	Two single wheels in tandem type landing gear (C130).
TRT	2T	Two triple wheels in tandem type landing gear (C17), etc.
DT	2D	Two dual wheels in tandem type landing gear (B707), etc.
TT	2D	Two dual wheels in tandem type landing gear (B757, KC135).
SBTT	2D/D1	Two dual wheels in tandem/dual wheel body gear type landing gear (KC10).
None	2D/2D1	Two dual wheels in tandem/two dual wheels in tandem body gear type landing gear (A340–600).
DDT	2D/2D2	Two dual wheels in tandem/two dual wheels in double tandem body gear type landing gear (B747, E4).
TTT	3D	Three dual wheels in tandem type landing gear (B777), etc.
TT	D2	Dual wheel gear two struts per side main gear type landing gear (B52).
TDT	C5	Complex dual wheel and quadruple wheel combination landing gear (C5).

AUW—All up weight. Maximum weight bearing capacity for any aircraft irrespective of landing gear configuration.

SWL—Single Wheel Loading. (This includes information submitted in terms of Equivalent Single Wheel Loading (ESWL) and Single Isolated Wheel Loading).

PSI—Pounds per square inch. PSI is the actual figure expressing maximum pounds per square inch runway will support, e.g., (SWL 000/PSI 535).

Omission of weight bearing capacity indicates information unknown.

The ACN/PCN System is the ICAO standard method of reporting pavement strength for pavements with bearing strengths greater than 12,500 pounds. The Pavement Classification Number (PCN) is established by an engineering assessment of the runway. The PCN is for use in conjunction with an Aircraft Classification Number (ACN). Consult the Aircraft Flight Manual, Flight Information Handbook, or other appropriate source for ACN tables or charts. Currently, ACN data may not be available for all aircraft. If an ACN table or chart is available, the ACN can be calculated by taking into account the aircraft weight, the pavement type, and the subgrade category. For runways that have been evaluated under the ACN/PCN system, the PCN will be shown as a five-part code (e.g. PCN 80 R/B/W/T). Details of the coded format are as follows:

NOTE: Prior permission from the airport controlling authority is required when the ACN of the aircraft exceeds the published PCN or aircraft tire pressure exceeds the published limits.

- (1) The PCN NUMBER—The reported PCN indicates that an aircraft with an ACN equal or less than the reported PCN can operate on the pavement subject to any limitation on the tire pressure.
- (2) The type of pavement:
R — Rigid
F — Flexible
- (3) The pavement subgrade category:
A — High
B — Medium
C — Low
D — Ultra-low
- (4) The maximum tire pressure authorized for the pavement:
W — Unlimited, no pressure limit
X — High, limited to 254 psi (1.75 MPa)
Y — Medium, limited to 181 psi (1.25MPa)
Z — Low, limited to 73 psi (0.50 MPa)
- (5) Pavement evaluation method:
T — Technical evaluation
U — By experience of aircraft using the pavement

RUNWAY LIGHTING

Lights are in operation sunset to sunrise. Lighting available by prior arrangement only or operating part of the night and/or pilot controlled lighting with specific operating hours are indicated under airport or military remarks. At USN/USMC facilities lights are available only during airport hours of operation. Since obstructions are usually lighted, obstruction lighting is not included in this code. Unlighted obstructions on or surrounding an airport will be noted in airport or military remarks. Runway lights nonstandard (NSTD) are systems for which the light fixtures are not FAA approved L-800 series: color, intensity, or spacing does not meet FAA standards. Nonstandard runway lights, VASI, or any other system not listed below will be shown in airport remarks or military

LEGEND 7.—Airport/Facility Directory.

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AIRPORT/FACILITY DIRECTORY LEGEND

service. Temporary, emergency or limited runway edge lighting such as flares, smudge pots, lanterns or portable runway lights will also be shown in airport remarks or military service. Types of lighting are shown with the runway or runway end they serve.

NSTD—Light system fails to meet FAA standards.	SALS—Short Approach Lighting System.
LIRL—Low Intensity Runway Lights.	SALSF—Short Approach Lighting System with Sequenced Flashing Lights.
MIRL—Medium Intensity Runway Lights.	SSALS—Simplified Short Approach Lighting System.
HIRL—High Intensity Runway Lights.	SSALF—Simplified Short Approach Lighting System with Sequenced Flashing Lights.
RAIL—Runway Alignment Indicator Lights.	SSALR—Simplified Short Approach Lighting System with Runway Alignment Indicator Lights.
REIL—Runway End Identifier Lights.	ALSAF—High Intensity Approach Lighting System with Sequenced Flashing Lights.
CL—Centerline Lights.	ALSF1—High Intensity Approach Lighting System with Sequenced Flashing Lights, Category I, Configuration.
TDZL—Touchdown Zone Lights.	ALSF2—High Intensity Approach Lighting System with Sequenced Flashing Lights, Category II, Configuration.
ODALS—Omni Directional Approach Lighting System.	SF—Sequenced Flashing Lights.
AF OVRN—Air Force Overrun 1000' Standard Approach Lighting System.	OLS—Optical Landing System.
MALS—Medium Intensity Approach Lighting System.	WAVE—OFF.
MALSF—Medium Intensity Approach Lighting System with Sequenced Flashing Lights.	
MALSR—Medium Intensity Approach Lighting System with Runway Alignment Indicator Lights.	
RLLS—Runway Lead-in Light System	

NOTE: Civil ALSF2 may be operated as SSALR during favorable weather conditions. When runway edge lights are positioned more than 10 feet from the edge of the usable runway surface a remark will be added in the "Remarks" portion of the airport entry. This is applicable to Air Force, Air National Guard and Air Force Reserve Bases, and those joint use airfields on which they are tenants.

VISUAL GLIDESLOPE INDICATORS

APAP—A system of panels, which may or may not be lighted, used for alignment of approach path.

PNIL	APAP on left side of runway	PNIR	APAP on right side of runway
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PAPI—Precision Approach Path Indicator

P2L	2-identical light units placed on left side of runway	P4L	4-identical light units placed on left side of runway
P2R	2-identical light units placed on right side of runway	P4R	4-identical light units placed on right side of runway

PVASI—Pulsating/steady burning visual approach slope indicator, normally a single light unit projecting two colors.

PSIL	PVASI on left side of runway	PSIR	PVASI on right side of runway
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SAVASI—Simplified Abbreviated Visual Approach Slope Indicator

S2L	2-box SAVASI on left side of runway	S2R	2-box SAVASI on right side of runway
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TRCV—Tri-color visual approach slope indicator, normally a single light unit projecting three colors.

TRIL	TRCV on left side of runway	TRIR	TRCV on right side of runway
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VASI—Visual Approach Slope Indicator

V2L	2-box VASI on left side of runway	V6L	6-box VASI on left side of runway
V2R	2-box VASI on right side of runway	V6R	6-box VASI on right side of runway
V4L	4-box VASI on left side of runway	V12	12-box VASI on both sides of runway
V4R	4-box VASI on right side of runway	V16	16-box VASI on both sides of runway

NOTE: Approach slope angle and threshold crossing height will be shown when available; i.e., -GA 3.5° TCH 37'.

PILOT CONTROL OF AIRPORT LIGHTING

Key Mike	Function
7 times within 5 seconds	Highest intensity available
5 times within 5 seconds	Medium or lower intensity (Lower REIL or REIL-Off)
3 times within 5 seconds	Lowest intensity available (Lower REIL or REIL-Off)

Available systems will be indicated in the Service section, e.g., LGT ACTIVATE HIRL Rwy 07-25, MALSR Rwy 07, and VASI Rwy 07-122.8.

Where the airport is not served by an instrument approach procedure and/or has an independent type system of different specification installed by the airport sponsor, descriptions of the type lights, method of control, and operating frequency will be explained in clear text. See AIM, "Basic Flight Information and ATC Procedures," for detailed description of pilot control of airport lighting.

RUNWAY SLOPE

When available, runway slope data will be provided. Runway slope will be shown only when it is 0.3 percent or greater. On runways less than 8000 feet, the direction of the slope up will be indicated, e.g., 0.3% up NW. On runways 8000 feet or greater, the slope will be shown (up or down) on the runway end line, e.g., RWY 13: 0.3% up., RWY 31: Pole. Rgt t/c. 0.4% down.

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RUNWAY END DATA

Information pertaining to the runway approach end such as approach lights, touchdown zone lights, runway end identification lights, visual glideslope indicators, displaced thresholds, controlling obstruction, and right hand traffic pattern, will be shown on the specific runway end. "Rgt ttc"—Right traffic indicates right turns should be made on landing and takeoff for specified runway end. Runway Visual Range shall be shown as "RVR" appended with "T" for touchdown, "M" for midpoint, and "R" for rollout; e.g., RVR-TMR.

(19) LAND AND HOLD—SHORT OPERATIONS (LAHSO)

LAHSO is an acronym for "Land and Hold—Short Operations." These operations include landing and holding short of an intersection runway, an intersecting taxiway, or other predetermined points on the runway other than a runway or taxiway. Measured distance represents the available landing distance on the landing runway, in feet.

Specific questions regarding these distances should be referred to the air traffic manager of the facility concerned. The Aeronautical Information Manual contains specific details on hold—short operations and markings.

(20) RUNWAY DECLARED DISTANCE INFORMATION

TORA—Take-off Run Available. The length of runway declared available and suitable for the ground run of an aeroplane take-off.

TODA—Take-off Distance Available. The length of the take-off run available plus the length of the clearway, if provided.

ASDA—Accelerate-Stop Distance Available. The length of the take-off run available plus the length of the stopway, if provided.

LDA—Landing Distance Available. The length of runway which is declared available and suitable for the ground run of an aeroplane landing.

(21) ARRESTING GEAR/SYSTEMS

Arresting gear is shown as it is located on the runway. The a-gear distance from the end of the appropriate runway (or into the overrun) is indicated in parentheses. A-Gear which has a bi-direction capability and can be utilized for emergency approach end engagement is indicated by a (B). Up to 15 minutes advance notice may be required for rigging A-Gear for approach and engagement. Airport listing may show availability of other than US Systems. This information is provided for emergency requirements only. Refer to current aircraft operating manuals for specific engagement weight and speed criteria based on aircraft structural restrictions and arresting system limitations.

Following is a list of current systems referenced in this publication identified by both Air Force and Navy terminology:

BI-DIRECTIONAL CABLE (B)

TYPE	DESCRIPTION
BAK-9	Rotary friction brake.
BAK-12A	Standard BAK-12 with 950 foot run out, 1-inch cable and 40,000 pound weight setting. Rotary friction brake.
BAK-12B	Extended BAK-12 with 1200 foot run, 1 ¼ inch Cable and 50,000 pounds weight setting. Rotary friction brake.
E28	Rotary Hydraulic (Water Brake).
M21	Rotary Hydraulic (Water Brake) Mobile.

The following device is used in conjunction with some aircraft arresting systems:

BAK-14	A device that raises a hook cable out of a slot in the runway surface and is remotely positioned for engagement by the tower on request. (In addition to personnel reaction time, the system requires up to five seconds to fully raise the cable.)
H	A device that raises a hook cable out of a slot in the runway surface and is remotely positioned for engagement by the tower on request. (In addition to personnel reaction time, the system requires up to one and one-half seconds to fully raise the cable.)

UNI-DIRECTIONAL CABLE

TYPE	DESCRIPTION
MB60	Textile brake—an emergency one-time use, modular braking system employing the tearing of specially woven textile straps to absorb the kinetic energy.
E5/E5-1/E5-3	Chain Type. At USN/USMC stations E-5 A-GEAR systems are rated, e.g., E-5 RATING-13R-1100 HW (DRY), 31L/R-1200 STD (WET). This rating is a function of the A-GEAR chain weight and length and is used to determine the maximum aircraft engaging speed. A dry rating applies to a stabilized surface (dry or wet) while a wet rating takes into account the amount (if any) of wet overrun that is not capable of withstanding the aircraft weight. These ratings are published under Service/Military/A-Gear in the entry.

FOREIGN CABLE

TYPE	DESCRIPTION	US EQUIVALENT
44B-3H	Rotary Hydraulic (Water Brake)	
CHAG	Chain	E-5

UNI-DIRECTIONAL BARRIER

TYPE	DESCRIPTION
MA-1A	Web barrier between stanchions attached to a chain energy absorber.
BAK-15	Web barrier between stanchions attached to an energy absorber (water squeezer, rotary friction, chain). Designed for wing engagement.

NOTE: Landing short of the runway threshold on a runway with a BAK-15 in the underrun is a significant hazard. The barrier in the down position still protrudes several inches above the underrun. Aircraft contact with the barrier short of the runway threshold can cause damage to the barrier and substantial damage to the aircraft.

OTHER

TYPE	DESCRIPTION
EMAS	Engineered Material Arresting System, located beyond the departure end of the runway, consisting of high energy absorbing materials which will crush under the weight of an aircraft.

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22 SERVICE**SERVICING—CIVIL**

- S1: Minor airframe repairs. S5: Major airframe repairs.
 S2: Minor airframe and minor powerplant repairs. S6: Minor airframe and major powerplant repairs.
 S3: Major airframe and minor powerplant repairs. S7: Major powerplant repairs.
 S4: Major airframe and major powerplant repairs. S8: Minor powerplant repairs.

FUEL—CIVIL

CODE	FUEL	CODE	FUEL
80	Grade 80 gasoline (Red)	A1+	Jet A-1, Kerosene with FS-II*, FP** minus 47° C.
100	Grade 100 gasoline (Green)	B	Jet B, Wide-cut, turbine fuel without FS-II*, FP** minus 50° C.
100LL	100LL gasoline (low lead) (Blue)	B+	Jet B, Wide-cut, turbine fuel with FS-II*, FP** minus 50° C
115	Grade 115 gasoline (115/145 military specification) (Purple)	J4 (JP4)	(JP-4 military specification) FP** minus 58° C.
A	Jet A, Kerosene, without FS-II*, FP** minus 40° C.	J5 (JP5)	(JP-5 military specification) Kerosene with FS-II, FP** minus 46°C.
A+	Jet A, Kerosene, with FS-II*, FP** minus 40°C.	J8 (JP8)	(JP-8 military specification) Jet A-1, Kerosene with FS-II*, Cl/LI#, SDA##, FP** minus 47°C.
A++	Jet A, Kerosene, with FS-II*, Cl/LI#, SDA##, FP** minus 40°C.	J8+100	(JP-8 military specification) Jet A-1, Kerosene with FS-II*, Cl/LI#, SDA##, FP** minus 47°C, with +100 fuel additive that improves thermal stability characteristics of kerosene jet fuels.
A++100	Jet A, Kerosene, with FS-II*, Cl/LI#, SDA##, FP** minus 40°C, with +100 fuel additive that improves thermal stability characteristics of kerosene jet fuels.	J	(Jet Fuel Type Unknown)
A1	Jet A-1, Kerosene, without FS-II*, FP** minus 47°C.	MOGAS	Automobile gasoline which is to be used as aircraft fuel.

*(Fuel System Icing Inhibitor) ** (Freeze Point) # (Corrosion Inhibitors/Lubricity Improvers) ## (Static Dissipator Additive)

NOTE: Certain automobile gasoline may be used in specific aircraft engines if a FAA supplemental type certificate has been obtained. Automobile gasoline, which is to be used in aircraft engines, will be identified as "MOGAS", however, the grade/type and other octane rating will not be published.

Data shown on fuel availability represents the most recent information the publisher has been able to acquire. Because of a variety of factors, the fuel listed may not always be obtainable by transient civil pilots. Confirmation of availability of fuel should be made directly with fuel suppliers at locations where refueling is planned.

OXYGEN—CIVIL

- OX 1 High Pressure OX 3 High Pressure—Replacement Bottles
 OX 2 Low Pressure OX 4 Low Pressure—Replacement Bottles

SERVICE—MILITARY

Specific military services available at the airport are listed under this general heading. Remarks applicable to any military service are shown in the individual service listing.

JET AIRCRAFT STARTING UNITS (JASU)—MILITARY

The numeral preceding the type of unit indicates the number of units available. The absence of the numeral indicates ten or more units available. If the number of units is unknown, the number one will be shown. Absence of JASU designation indicates non-availability.

The following is a list of current JASU systems referenced in this publication:

USAF JASU (For variations in technical data, refer to T.O. 35-1-7.)**ELECTRICAL STARTING UNITS:**

A/M32A-86	AC: 115/200v, 3 phase, 90 kva, 0.8 pf, 4 wire DC: 28v, 1500 amp, 72 kw (with TR pack)
MC-1A	AC: 115/208v, 400 cycle, 3 phase, 37.5 kva, 0.8 pf, 108 amp, 4 wire DC: 28v, 500 amp, 14 kw
MD-3	AC: 115/208v, 400 cycle, 3 phase, 60 kva, 0.75 pf, 4 wire DC: 28v, 1500 amp, 45 kw, split bus
MD-3A	AC: 115/208v, 400 cycle, 3 phase, 60 kva, 0.75 pf, 4 wire DC: 28v, 1500 amp, 45 kw, split bus
MD-3M	AC: 115/208v, 400 cycle, 3 phase, 60 kva, 0.75 pf, 4 wire DC: 28v, 500 amp, 15 kw
MD-4	AC: 120/208v, 400 cycle, 3 phase, 62.5 kva, 0.8 pf, 175 amp, "WYE" neutral ground, 4 wire, 120v, 400 cycle, 3 phase, 62.5 kva, 0.8 pf, 303 amp, "DELTA" 3 wire, 120v, 400 cycle, 1 phase, 62.5 kva, 0.8 pf, 520 amp, 2 wire

AIRPORT/FACILITY DIRECTORY LEGEND**21****AIR STARTING UNITS**

AM32-95	150 +/- 5 lb/min (2055 +/- 68 cfm) at 51 +/- 2 psia
AM32A-95	150 +/- 5 lb/min @ 49 +/- 2 psia (35 +/- 2 psig)
LASS	150 +/- 5 lb/min @ 49 +/- 2 psia
MA-1A	82 lb/min (1123 cfm) at 130° air inlet temp, 45 psia (min) air outlet press
MC-1	15 cfm, 3500 psia
MC-1A	15 cfm, 3500 psia
MC-2A	15 cfm, 200 psia
MC-11	8,000 cu in cap, 4000 psig, 15 cfm

COMBINED AIR AND ELECTRICAL STARTING UNITS:

AGPU	AC: 115/200v, 400 cycle, 3 phase, 30 kw gen DC: 28v, 700 amp AIR: 60 lb/min @ 40 psig @ sea level
AM32A-60*	AIR: 120 +/- 4 lb/min (1644 +/- 55 cfm) at 49 +/- 2 psia AC: 120/208v, 400 cycle, 3 phase, 75 kva, 0.75 pf, 4 wire, 120v, 1 phase, 25 kva DC: 28v, 500 amp, 15 kw
AM32A-60A	AIR: 150 +/- 5 lb/min (2055 +/- 68 cfm at 51 +/- psia AC: 120/208v, 400 cycle, 3 phase, 75 kva, 0.75 pf, 4 wire DC: 28v, 200 amp, 5.6 kw
AM32A-60B*	AIR: 130 lb/min, 50 psia AC: 120/208v, 400 cycle, 3 phase, 75 kva, 0.75 pf, 4 wire DC: 28v, 200 amp, 5.6 kw

*NOTE: During combined air and electrical loads, the pneumatic circuitry takes preference and will limit the amount of electrical power available.

USN JASU**ELECTRICAL STARTING UNITS:**

NC-8A/A1	DC: 500 amp constant, 750 amp intermittent, 28v; AC: 60 kva @ .8 pf, 115/200v, 3 phase, 400 Hz.
NC-10A/A1/B/C	DC: 750 amp constant, 1000 amp intermittent, 28v; AC: 90 kva, 115/200v, 3 phase, 400 Hz.

AIR STARTING UNITS:

GTC-85/GTE-85	120 lbs/min @ 45 psi.
MSU-200NAV/A/U47A-5	204 lbs/min @ 56 psia.
WELLS AIR START SYSTEM	180 lbs/min @ 75 psi or 120 lbs/min @ 45 psi. Simultaneous multiple start capability.

COMBINED AIR AND ELECTRICAL STARTING UNITS:

NCPP-105/RCPT	180 lbs/min @ 75 psi or 120 lbs/min @ 45 psi. 700 amp, 28v DC. 120/208v, 400 Hz AC, 30 kva.
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ARMY JASU

59B2-1B	28v, 7.5 kw, 280 amp.
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OTHER JASU**ELECTRICAL STARTING UNITS (DND):**

CE12	AC 115/200v, 140 kva, 400 Hz, 3 phase
CE13	AC 115/200v, 60 kva, 400 Hz, 3 phase
CE14	AC/DC 115/200v, 140 kva, 400 Hz, 3 phase, 28vDC, 1500 amp
CE15	DC 22-35v, 500 amp continuous 1100 amp intermittent
CE16	DC 22-35v, 500 amp continuous 1100 amp intermittent soft start

AIR STARTING UNITS (DND):

CA2	ASA 45.5 psig, 116.4 lb/min
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COMBINED AIR AND ELECTRICAL STARTING UNITS (DND)

CEA1	AC 120/208v, 60 kva, 400 Hz, 3 phase DC 28v, 75 amp AIR 112.5 lb/min, 47 psig
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ELECTRICAL STARTING UNITS (OTHER)

C-26	28v 45kw 115-200v 15kw 380-800 Hz 1 phase 2 wire
C-26-B, C-26-C	28v 45kw: Split Bus: 115-200v 15kw 380-800 Hz 1 phase 2 wire
E3	DC 28v/10kw

AIR STARTING UNITS (OTHER):

A4	40 psi/2 lb/sec (LPAS Mk12, Mk12L, Mk12A, Mk1, Mk2B)
MA-1	150 Air HP, 115 lb/min 50 psia
MA-2	250 Air HP, 150 lb/min 75 psia

CARTRIDGE:

MXU-4A	USAF
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LEGEND 11.—Airport/Facility Directory.

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AIRPORT/FACILITY DIRECTORY LEGEND

FUEL—MILITARY

Fuel available through US Military Base supply, DESC Into-Plane Contracts and/or reciprocal agreement is listed first and is followed by (Mil). At commercial airports where Into-Plane contracts are in place, the name of the refueling agent is shown. Military fuel should be used first if it is available. When military fuel cannot be obtained but Into-Plane contract fuel is available, Government aircraft must refuel with the contract fuel and applicable refueling agent to avoid any breach in contract terms and conditions. Fuel not available through the above is shown preceded by NC (no contract). When fuel is obtained from NC sources, local purchase procedures must be followed. The US Military Aircraft Identaplates DD Form 1896 (Jet Fuel), DD Form 1897 (Avgas) and AF Form 1245 (Avgas) are used at military installations only. The US Government Aviation Into-Plane Reimbursement (AIR) Card (currently issued by AVCARD) is the instrument to be used to obtain fuel under a DESC Into-Plane Contract and for NC purchases if the refueling agent at the commercial airport accepts the AVCARD. A current list of contract fuel locations is available online at https://cis.energy.dla.mil/ip_cis/. See legend item 14 for fuel code and description.

SUPPORTING FLUIDS AND SYSTEMS—MILITARY

CODE

ADI	Anti-Detonation Injection Fluid—Reciprocating Engine Aircraft.
W	Water Thrust Augmentation—Jet Aircraft.
WAI	Water-Alcohol Injection Type, Thrust Augmentation—Jet Aircraft.
SP	Single Point Refueling.
PRESAIR	Air Compressors rated 3,000 PSI or more.
De-Ice	Anti-icing/De-icing/Defrosting Fluid (MIL-A-8243).

OXYGEN:

LPOX	Low pressure oxygen servicing.
HPOX	High pressure oxygen servicing.
LHOX	Low and high pressure oxygen servicing.
LOX	Liquid oxygen servicing.
OXR	Oxygen replacement bottles. (Maintained primarily at Naval stations for use in acft where oxygen can be replenished only by replacement of cylinders.)
OX	Indicates oxygen servicing when type of servicing is unknown.

NOTE: Combinations of above items is used to indicate complete oxygen servicing available;

LHOXR	Low and high pressure oxygen servicing and replacement bottles;
LPOXR	Low pressure oxygen replacement bottles only, etc.

NOTE: Aircraft will be serviced with oxygen procured under military specifications only. Aircraft will not be serviced with medical oxygen.

NITROGEN:

LPNIT	Low pressure nitrogen servicing.
HPNIT	High pressure nitrogen servicing.
LHNIT	Low and high pressure nitrogen servicing.

OIL—MILITARY

US AVIATION OILS (MIL SPECS):

CODE	GRADE, TYPE
O-113	1065, Reciprocating Engine Oil (MIL-L-6082)
O-117	1100, Reciprocating Engine Oil (MIL-L-6082)
O-117+	1100, O-117 plus cyclohexanone (MIL-L-6082)
O-123	1065, (Dispersant), Reciprocating Engine Oil (MIL-L-22851 Type III)
O-128	1100, (Dispersant), Reciprocating Engine Oil (MIL-L-22851 Type II)
O-132	1005, Jet Engine Oil (MIL-L-6081)
O-133	1010, Jet Engine Oil (MIL-L-6081)
O-147	None, MIL-L-6085A Lubricating Oil, Instrument, Synthetic
O-148	None, MIL-L-7808 (Synthetic Base) Turbine Engine Oil
O-149	None, Aircraft Turbine Engine Synthetic, 7.5c St
O-155	None, MIL-L-6086C, Aircraft, Medium Grade
O-156	None, MIL-L-23699 (Synthetic Base), Turboprop and Turboshaft Engines
JOAP/SOAP	Joint Oil Analysis Program. JOAP support is furnished during normal duty hours, other times on request. (JOAP and SOAP programs provide essentially the same service, JOAP is now the standard joint service supported program.)

TRANSIENT ALERT (TRAN ALERT)—MILITARY

Tran Alert service is considered to include all services required for normal aircraft turn-around, e.g., servicing (fuel, oil, oxygen, etc.), debriefing to determine requirements for maintenance, minor maintenance, inspection and parking assistance of transient aircraft. Drag chute repack, specialized maintenance, or extensive repairs will be provided within the capabilities and priorities of the base. Delays can be anticipated after normal duty hours/holidays/weekends regardless of the hours of transient maintenance operation. Pilots should not expect aircraft to be serviced for TURN-AROUNDS during time periods when servicing or maintenance manpower is not available. In the case of airports not operated exclusively by US military, the servicing indicated by the remarks will not always be available for US military aircraft. When transient alert services are not shown, facilities are unknown. NO PRIORITY BASIS—means that transient alert services will be provided only after all the requirements for mission/tactical assigned aircraft have been accomplished.

AIRPORT/FACILITY DIRECTORY LEGEND

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(23) AIRPORT REMARKS

The Attendance Schedule is the months, days and hours the airport is actually attended. Airport attendance does not mean watchman duties or telephone accessibility, but rather an attendant or operator on duty to provide at least minimum services (e.g., repairs, fuel, transportation).

Airport Remarks have been grouped in order of applicability. Airport remarks are limited to those items of information that are determined essential for operational use, i.e., conditions of a permanent or indefinite nature and conditions that will remain in effect for more than 30 days concerning aeronautical facilities, services, maintenance available, procedures or hazards, knowledge of which is essential for safe and efficient operation of aircraft. Information concerning permanent closing of a runway or taxiway will not be shown. A note "See Special Notices" shall be applied within this remarks section when a special notice applicable to the entry is contained in the Special Notices section of this publication.

Parachute Jumping indicates parachute jumping areas associated with the airport. See Parachute Jumping Area section of this publication for additional Information.

Landing Fee indicates landing charges for private or non-revenue producing aircraft. In addition, fees may be charged for planes that remain over a couple of hours and buy no services, or at major airline terminals for all aircraft.

Note: Unless otherwise stated, remarks including runway ends refer to the runway's approach end.

(24) MILITARY REMARKS

Joint Civil/Military airports contain both Airport Remarks and Military Remarks. Military Remarks published for these airports are applicable only to the military. Military and joint Military/Civil airports contain only Military Remarks. Remarks contained in this section may not be applicable to civil users. When both sets of remarks exist, the first set is applicable to the primary operator of the airport. Remarks applicable to a tenant on the airport are shown preceded by the tenant organization, i.e., (A) (AF) (N) (ANG), etc. Military airports operate 24 hours unless otherwise specified. Airport operating hours are listed first (airport operating hours will only be listed if they are different than the airport attended hours or if the attended hours are unavailable) followed by pertinent remarks in order of applicability. Remarks will include information on restrictions, hazards, traffic pattern, noise abatement, customs/agriculture/immigration, and miscellaneous information applicable to the Military.

Type of restrictions:

CLOSED: When designated closed, the airport is restricted from use by all aircraft unless stated otherwise. Any closure applying to specific type of aircraft or operation will be so stated. USN/USMC/USAF airports are considered closed during non-operating hours. Closed airports may be utilized during an emergency provided there is a safe landing area.

OFFICIAL BUSINESS ONLY: The airfield is closed to all transient military aircraft for obtaining routine services such as fueling, passenger drop off or pickup, practice approaches, parking, etc. The airfield may be used by aircrews and aircraft if official government business (including civilian) must be conducted on or near the airfield and prior permission is received from the airfield manager.

AF OFFICIAL BUSINESS ONLY OR NAVY OFFICIAL BUSINESS ONLY: Indicates that the restriction applies only to service indicated.

PRIOR PERMISSION REQUIRED (PPR): Airport is closed to transient aircraft unless approval for operation is obtained from the appropriate commander through Chief, Airfield Management or Airfield Operations Officer. Official Business or PPR does not preclude the use of US Military airports as an alternate for IFR flights. If a non-US military airport is used as a weather alternate and requires a PPR, the PPR must be requested and confirmed before the flight departs. The purpose of PPR is to control volume and flow of traffic rather than to prohibit it. Prior permission is required for all aircraft requiring transient alert service outside the published transient alert duty hours. All aircraft carrying hazardous materials must obtain prior permission as outlined in AFJI 11-204, AR 95-27, OPNAVINST 3710.7.

Note: OFFICIAL BUSINESS ONLY AND PPR restrictions are not applicable to Special Air Mission (SAM) or Special Air Resource (SPAR) aircraft providing person or persons on board are designated Code 6 or higher as explained in AFJMAN 11-213, AR 95-11, OPNAVINST 3722-8J. Official Business Only or PPR do not preclude the use of the airport as an alternate for IFR flights.

(25) AIRPORT MANAGER

The phone number of the airport manager.

(26) WEATHER DATA SOURCES

Weather data sources will be listed alphabetically followed by their assigned frequencies and/or telephone number and hours of operation.

ASOS—Automated Surface Observing System. Reports the same as an AWOS-3 plus precipitation identification and intensity, and freezing rain occurrence;

AWOS—Automated Weather Observing System

AWOS-A—reports altimeter setting (all other information is advisory only).

AWOS-AV—reports altimeter and visibility.

AWOS-1—reports altimeter setting, wind data and usually temperature, dew point and density altitude.

AWOS-2—reports the same as AWOS-1 plus visibility.

AWOS-3—reports the same as AWOS-1 plus visibility and cloud/ceiling data.

AWOS-3P reports the same as the AWOS-3 system, plus a precipitation identification sensor.

AWOS-3PT reports the same as the AWOS-3 system, plus precipitation identification sensor and a thunderstorm/lightning reporting capability.

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AIRPORT/FACILITY DIRECTORY LEGEND

AWOS-3T reports the same as AWOS-3 system and includes a thunderstorm/lightning reporting capability.

See AIM, Basic Flight Information and ATC Procedures for detailed description of Weather Data Sources.

AWOS-4—reports same as AWOS-3 system, plus precipitation occurrence, type and accumulation, freezing rain, thunderstorm and runway surface sensors.

HIWAS—See RADIO AIDS TO NAVIGATION

LAWRS—Limited Aviation Weather Reporting Station where observers report cloud height, weather, obstructions to vision, temperature and dewpoint (in most cases), surface wind, altimeter and pertinent remarks.

LLWAS—indicates a Low Level Wind Shear Alert System consisting of a center field and several field perimeter anemometers.

SAWRS—identifies airports that have a Supplemental Aviation Weather Reporting Station available to pilots for current weather information.

SWSL—Supplemental Weather Service Location providing current local weather information via radio and telephone.

TDWR—indicates airports that have Terminal Doppler Weather Radar.

WSP—indicates airports that have Weather System Processor.

When the automated weather source is broadcast over an associated airport NAVAID frequency (see NAVAID line), it shall be indicated by a bold ASOS, AWOS, or HIWAS followed by the frequency, identifier and phone number, if available.

27 COMMUNICATIONS

Airport terminal control facilities and radio communications associated with the airport shall be shown. When the call sign is not the same as the airport name the call sign will be shown. Frequencies shall normally be shown in descending order with the primary frequency listed first. Frequencies will be listed, together with sectorization indicated by outbound radials, and hours of operation. Communications will be listed in sequence as follows:

Single Frequency Approach (SFA), Common Traffic Advisory Frequency (CTAF), Aeronautical Advisory Stations (UNICOM) or (AUNICOM), and Automatic Terminal Information Service (ATIS) along with their frequency is shown, where available, on the line following the heading "COMMUNICATIONS." When the CTAF and UNICOM frequencies are the same, the frequency will be shown as CTAF/UNICOM 122.8.

The FSS telephone nationwide is toll free 1-800-WX-BRIEF (1-800-992-7433). When the FSS is located on the field it will be indicated as "on arpt". Frequencies available at the FSS will follow in descending order. Remote Communications Outlet (RCO) providing service to the airport followed by the frequency and FSS RADIO name will be shown when available. FSS's provide information on airport conditions, radio aids and other facilities, and process flight plans. Airport Advisory Service (AAS) is provided on the CTAF by FSS's for select non-tower airports or airports where the tower is not in operation.

(See AIM, Para 4-1-9 Traffic Advisory Practices at Airports Without Operating Control Towers or AC 90-42C.)

Aviation weather briefing service is provided by FSS specialists. Flight and weather briefing services are also available by calling the telephone numbers listed.

Remote Communications Outlet (RCO)—An unmanned air/ground communications facility that is remotely controlled and provides UHF or VHF communications capability to extend the service range of an FSS.

Civil Communications Frequencies—Civil communications frequencies used in the FSS air/ground system are operated on 122.0, 122.2, 123.6; emergency 121.5; plus receive-only on 122.1.

- 122.0 is assigned as the Enroute Flight Advisory Service frequency at selected FSS RADIO outlets.
- 122.2 is assigned as a common enroute frequency.
- 123.6 is assigned as the airport advisory frequency at select non-tower locations. At airports with a tower, FSS may provide airport advisories on the tower frequency when tower is closed.
- 122.1 is the primary receive-only frequency at VOR's.
- Some FSS's are assigned 50 kHz frequencies in the 122-126 MHz band (eg. 122.45). Pilots using the FSS A/G system should refer to this directory or appropriate charts to determine frequencies available at the FSS or remoted facility through which they wish to communicate.

Emergency frequency 121.5 and 243.0 are available at all Flight Service Stations, most Towers, Approach Control and RADAR facilities.

Frequencies published followed by the letter "T" or "R", indicate that the facility will only transmit or receive respectively on that frequency. All radio aids to navigation (NAVAID) frequencies are transmit only.

TERMINAL SERVICES

SFA—Single Frequency Approach.

CTAF—A program designed to get all vehicles and aircraft at airports without an operating control tower on a common frequency.

ATIS—A continuous broadcast of recorded non-control information in selected terminal areas.

D-ATIS—Digital ATIS provides ATIS information in text form outside the standard reception range of conventional ATIS via landline & data link communications and voice message within range of existing transmitters.

AUNICOM—Automated UNICOM is a computerized, command response system that provides automated weather, radio check capability and airport advisory information selected from an automated menu by microphone clicks.

UNICOM—A non-government air/ground radio communications facility which may provide airport information.

PTD—Pilot to Dispatcher.

APP CON—Approach Control. The symbol **®** indicates radar approach control.

TOWER—Control tower.

GCA—Ground Control Approach System.

GND CON—Ground Control.

LEGEND 14.—Airport/Facility Directory.

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GCO—Ground Communication Outlet—An unstaffed, remotely controlled, ground/ground communications facility. Pilots at uncontrolled airports may contact ATC and FSS via VHF to a telephone connection to obtain an instrument clearance or close a VFR or IFR flight plan. They may also get an updated weather briefing prior to takeoff. Pilots will use four “key clicks” on the VHF radio to contact the appropriate ATC facility or six “key clicks” to contact the FSS. The GCO system is intended to be used only on the ground.

DEP CON—Departure Control. The symbol **Ⓡ** indicates radar departure control.

CLNC DEL—Clearance Delivery.

CPDLC—Controller Pilot Data Link Communication. FANS ATC data communication capability from the aircraft to the ATC Data Link system.

PRE TAXI CLNC—Pre taxi clearance.

VFR ADVSY SVC—VFR Advisory Service. Service provided by Non-Radar Approach Control.

Advisory Service for VFR aircraft (upon a workload basis) ctc APP CON.

COMD POST—Command Post followed by the operator call sign in parenthesis.

PMSV—Pilot-to-Metro Service call sign, frequency and hours of operation, when full service is other than continuous. PMSV installations at which weather observation service is available shall be indicated, following the frequency and/or hours of operation as “Wx obsn svc 1900–0000Z+” or “other times” may be used when no specific time is given. PMSV facilities manned by forecasters are considered “Full Service”. PMSV facilities manned by weather observers are listed as “Limited Service”.

OPS—Operations followed by the operator call sign in parenthesis.

CON

RANGE

FLT FLW—Flight Following

MEDIVAC

NOTE: Communication frequencies followed by the letter “X” indicate frequency available on request.

(28) AIRSPACE

Information concerning Class B, C, and part-time D and E surface area airspace shall be published with effective times, if available.

CLASS B—Radar Sequencing and Separation Service for all aircraft in CLASS B airspace.

CLASS C—Separation between IFR and VFR aircraft and sequencing of VFR arrivals to the primary airport.

TRSA—Radar Sequencing and Separation Service for participating VFR Aircraft within a Terminal Radar Service Area.

Class C, D, and E airspace described in this publication is that airspace usually consisting of a 5 NM radius core surface area that begins at the surface and extends upward to an altitude above the airport elevation (charted in MSL for Class C and Class D).

Class E surface airspace normally extends from the surface up to but not including the overlying controlled airspace.

When part-time Class C or Class D airspace defaults to Class E, the core surface area becomes Class E. This will be formatted as:

AIRSPACE: CLASS C svc “times” ctc **APP CON** other times CLASS E:

or

AIRSPACE: CLASS D svc “times” other times CLASS E.

When a part-time Class C, Class D or Class E surface area defaults to Class G, the core surface area becomes Class G up to, but not including, the overlying controlled airspace. Normally, the overlying controlled airspace is Class E airspace beginning at either 700’ or 1200’ AGL and may be determined by consulting the relevant VFR Sectional or Terminal Area Charts. This will be formatted as:

AIRSPACE: CLASS C svc “times” ctc **APP CON** other times CLASS G, with CLASS E 700’ (or 1200’) AGL & abv:

or

AIRSPACE: CLASS D svc “times” other times CLASS G with CLASS E 700’ (or 1200’) AGL & abv:

or

AIRSPACE: CLASS E svc “times” other times CLASS G with CLASS E 700’ (or 1200’) AGL & abv.

NOTE: AIRSPACE SVC “TIMES” INCLUDE ALL ASSOCIATED ARRIVAL EXTENSIONS. Surface area arrival extensions for instrument approach procedures become part of the primary core surface area. These extensions may be either Class D or Class E airspace and are effective concurrent with the times of the primary core surface area. For example, when a part-time Class C, Class D or Class E surface area defaults to Class G, the associated arrival extensions will default to Class G at the same time. When a part-time Class C or Class D surface area defaults to Class E, the arrival extensions will remain in effect as Class E airspace.

NOTE: CLASS E AIRSPACE EXTENDING UPWARD FROM 700 FEET OR MORE ABOVE THE SURFACE, DESIGNATED IN CONJUNCTION WITH AN AIRPORT WITH AN APPROVED INSTRUMENT PROCEDURE.

Class E 700’ AGL (shown as magenta vignette on sectional charts) and 1200’ AGL (blue vignette) areas are designated when necessary to provide controlled airspace for transitioning to/from the terminal and enroute environments. Unless otherwise specified, these 700’/1200’ AGL Class E airspace areas remain in effect continuously, regardless of airport operating hours or surface area status. These transition areas should not be confused with surface areas or arrival extensions.

(See Chapter 3, AIRSPACE, in the Aeronautical Information Manual for further details)

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AIRPORT/FACILITY DIRECTORY LEGEND

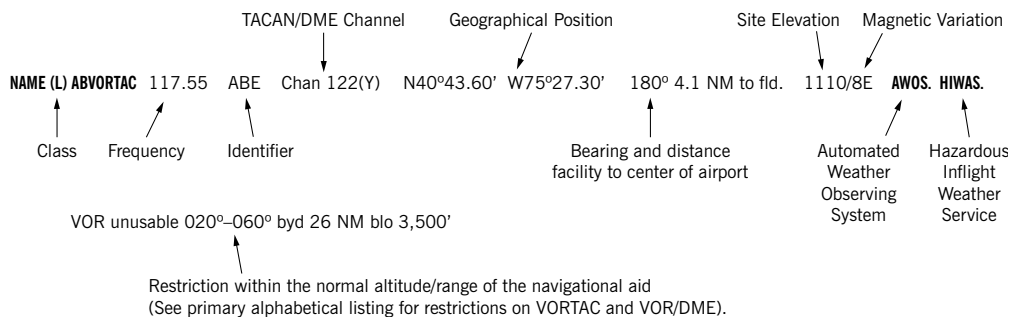
(29) VOR TEST FACILITY (VOT)

The VOT transmits a signal which provided users a convenient means to determine the operational status and accuracy of an aircraft VOR receiver while on the ground. Ground based VOTs and the associated frequency shall be shown when available. VOTs are also shown with identifier, frequency and referenced remarks in the VOR Receiver Check section in the back of this publication.

(30) RADIO AIDS TO NAVIGATION

The Airport/Facility Directory section of the Chart Supplement lists, by facility name, all Radio Aids to Navigation that appear on FAA, Aeronautical Information Services Visual or IFR Aeronautical Charts and those upon which the FAA has approved an Instrument Approach Procedure, with exception of selected TACANs. All VOR, VORTAC, TACAN and ILS equipment in the National Airspace System has an automatic monitoring and shutdown feature in the event of malfunction. Unmonitored, as used in this publication, for any navigational aid, means that monitoring personnel cannot observe the malfunction or shutdown signal. The NAVAID NOTAM file identifier will be shown as "NOTAM FILE IAD" and will be listed on the Radio Aids to Navigation line. When two or more NAVAIDS are listed and the NOTAM file identifier is different from that shown on the Radio Aids to Navigation line, it will be shown with the NAVAID listing. NOTAM file identifiers for ILSs and its components (e.g., NDB (LOM) are the same as the associated airports and are not repeated. Automated Surface Observing System (ASOS), Automated Weather Observing System (AWOS), and Hazardous Inflight Weather Advisory Service (HIWAS) will be shown when this service is broadcast over selected NAVAIDS.

NAVAID information is tabulated as indicated in the following sample:



Note: Those DME channel numbers with a (Y) suffix require TACAN to be placed in the "Y" mode to receive distance information.

HIWAS—Hazardous Inflight Weather Advisory Service is a continuous broadcast of inflight weather advisories including summarized SIGMETs, convective SIGMETs, AIRMETs and urgent PIREPs. HIWAS is presently broadcast over selected VOR's throughout the U.S.

ASR/PAR—Indicates that Surveillance (ASR) or Precision (PAR) radar instrument approach minimums are published in the U.S. Terminal Procedures. Only part-time hours of operation will be shown.

AIRPORT/FACILITY DIRECTORY LEGEND**27****RADIO CLASS DESIGNATIONS**

VOR/DME/TACAN Standard Service Volume (SSV) Classifications

<u>SSV Class</u>	<u>Altitudes</u>	<u>Distance (NM)</u>
(T) Terminal	1000' to 12,000'	25
(L) Low Altitude	1000' to 18,000'	40
(H) High Altitude	1000' to 14,500'	40
	14,500' to 18,000'	100
	18,000' to 45,000'	130
	45,000' to 60,000'	100

NOTE: Additionally, (H) facilities provide (L) and (T) service volume and (L) facilities provide (T) service. Altitudes are with respect to the station's site elevation. Coverage is not available in a cone of airspace directly above the facility.

The term VOR is, operationally, a general term covering the VHF omnidirectional bearing type of facility without regard to the fact that the power, the frequency protected service volume, the equipment configuration, and operational requirements may vary between facilities at different locations.

AB _____	Automatic Weather Broadcast.
DF _____	Direction Finding Service.
DME _____	UHF standard (TACAN compatible) distance measuring equipment.
DME(Y) _____	UHF standard (TACAN compatible) distance measuring equipment that require TACAN to be placed in the "Y" mode to receive DME.
GS _____	Glide slope.
H _____	Non-directional radio beacon (homing), power 50 watts to less than 2,000 watts (50 NM at all altitudes).
HH _____	Non-directional radio beacon (homing), power 2,000 watts or more (75 NM at all altitudes).
H-SAB _____	Non-directional radio beacons providing automatic transcribed weather service.
ILS _____	Instrument Landing System (voice, where available, on localizer channel).
IM _____	Inner marker.
LDA _____	Localizer Directional Aid.
LMM _____	Compass locator station when installed at middle marker site (15 NM at all altitudes).
LOM _____	Compass locator station when installed at outer marker site (15 NM at all altitudes).
MH _____	Non-directional radio beacon (homing) power less than 50 watts (25 NM at all altitudes).
MM _____	Middle marker.
OM _____	Outer marker.
S _____	Simultaneous range homing signal and/or voice.
SABH _____	Non-directional radio beacon not authorized for IFR or ATC. Provides automatic weather broadcasts.
SDF _____	Simplified Direction Facility.
TACAN _____	UHF navigational facility—omnidirectional course and distance information.
VOR _____	VHF navigational facility—omnidirectional course only.
VOR/DME _____	Collocated VOR navigational facility and UHF standard distance measuring equipment.
VORTAC _____	Collocated VOR and TACAN navigational facilities.
W _____	Without voice on radio facility frequency.
Z _____	VHF station location marker at a LF radio facility.

AIRPORT/FACILITY DIRECTORY LEGEND**ILS FACILITY PERFORMANCE CLASSIFICATION CODES**

Codes define the ability of an ILS to support autoland operations. The two portions of the code represent Official Category and farthest point along a Category I, II, or III approach that the Localizer meets Category III structure tolerances.

Official Category: I, II, or III; the lowest minima on published or unpublished procedures supported by the ILS.

Farthest point of satisfactory Category III Localizer performance for Category I, II, or III approaches: A – 4 NM prior to runway threshold, B – 3500 ft prior to runway threshold, C – glide angle dependent but generally 750–1000 ft prior to threshold, T – runway threshold, D – 3000 ft after runway threshold, and E – 2000 ft prior to stop end of runway.

ILS information is tabulated as indicated in the following sample:

ILS/DME 108.5 I-ORL Chan 22 Rwy 18. Class IIE. LOM HERNY NDB.

ILS Facility Performance
Classification Code

FREQUENCY PAIRING TABLE

VHF FREQUENCY	TACAN CHANNEL	VHF FREQUENCY	TACAN CHANNEL	VHF FREQUENCY	TACAN CHANNEL	VHF FREQUENCY	TACAN CHANNEL
108.10	18X	108.55	22Y	111.05	47Y	114.85	95Y
108.30	20X	108.65	23Y	111.15	48Y	114.95	96Y
108.50	22X	108.75	24Y	111.25	49Y	115.05	97Y
108.70	24X	108.85	25Y	111.35	50Y	115.15	98Y
108.90	26X	108.95	26Y	111.45	51Y	115.25	99Y
109.10	28X	109.05	27Y	111.55	52Y	115.35	100Y
109.30	30X	109.15	28Y	111.65	53Y	115.45	101Y
109.50	32X	109.25	29Y	111.75	54Y	115.55	102Y
109.70	34X	109.35	30Y	111.85	55Y	115.65	103Y
109.90	36X	109.45	31Y	111.95	56Y	115.75	104Y
110.10	38X	109.55	32Y	113.35	80Y	115.85	105Y
110.30	40X	109.65	33Y	113.45	81Y	115.95	106Y
110.50	42X	109.75	34Y	113.55	82Y	116.05	107Y
110.70	44X	109.85	35Y	113.65	83Y	116.15	108Y
110.90	46X	109.95	36Y	113.75	84Y	116.25	109Y
111.10	48X	110.05	37Y	113.85	85Y	116.35	110Y
111.30	50X	110.15	38Y	113.95	86Y	116.45	111Y
111.50	52X	110.25	39Y	114.05	87Y	116.55	112Y
111.70	54X	110.35	40Y	114.15	88Y	116.65	113Y
111.90	56X	110.45	41Y	114.25	89Y	116.75	114Y
108.05	17Y	110.55	42Y	114.35	90Y	116.85	115Y
108.15	18Y	110.65	43Y	114.45	91Y	116.95	116Y
108.25	19Y	110.75	44Y	114.55	92Y	117.05	117Y
108.35	20Y	110.85	45Y	114.65	93Y	117.15	118Y
108.45	21Y	110.95	46Y	114.75	94Y	117.25	119Y

FREQUENCY PAIRING TABLE

The following is a list of paired VOR/ILS VHF frequencies with TACAN channels.

TACAN CHANNEL	VHF FREQUENCY	TACAN CHANNEL	VHF FREQUENCY	TACAN CHANNEL	VHF FREQUENCY	TACAN CHANNEL	VHF FREQUENCY
2X	134.5	25X	108.80	36X	109.90	47X	111.00
2Y	134.55	25Y	108.85	36Y	109.95	47Y	111.05
11X	135.4	26X	108.90	37X	110.00	48X	111.10
11Y	135.45	26Y	108.95	37Y	110.05	48Y	111.15
12X	135.5	27X	109.00	38X	110.10	49X	111.20
12Y	135.55	27Y	109.05	38Y	110.15	49Y	111.25
17X	108.00	28X	109.10	39X	110.20	50X	111.30
17Y	108.05	28Y	109.15	39Y	110.25	50Y	111.35
18X	108.10	29X	109.20	40X	110.30	51X	111.40
18Y	108.15	29Y	109.25	40Y	110.35	51Y	111.45
19X	108.20	30X	109.30	41X	110.40	52X	111.50
19Y	108.25	30Y	109.35	41Y	110.45	52Y	111.55
20X	108.30	31X	109.40	42X	110.50	53X	111.60
20Y	108.35	31Y	109.45	42Y	110.55	53Y	111.65
21X	108.40	32X	109.50	43X	110.60	54X	111.70
21Y	108.45	32Y	109.55	43Y	110.65	54Y	111.75
22X	108.50	33X	109.60	44X	110.70	55X	111.80
22Y	108.55	33Y	109.65	44Y	110.75	55Y	111.85
23X	108.60	34X	109.70	45X	110.80	56X	111.90
23Y	108.65	34Y	109.75	45Y	110.85	56Y	111.95
24X	108.70	35X	109.80	46X	110.90	57X	112.00
24Y	108.75	35Y	109.85	46Y	110.95	57Y	112.05

AIRPORT/FACILITY DIRECTORY LEGEND

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TACAN CHANNEL	VHF FREQUENCY	TACAN CHANNEL	VHF FREQUENCY	TACAN CHANNEL	VHF FREQUENCY	TACAN CHANNEL	VHF FREQUENCY
58X	112.10	77X	113.00	96X	114.90	115X	116.80
58Y	112.15	77Y	113.05	96Y	114.95	115Y	116.85
59X	112.20	78X	113.10	97X	115.00	116X	116.90
59Y	112.25	78Y	113.15	97Y	115.05	116Y	116.95
60X	133.30	79X	113.20	98X	115.10	117X	117.00
60Y	133.35	79Y	113.25	98Y	115.15	117Y	117.05
61X	133.40	80X	113.30	99X	115.20	118X	117.10
61Y	133.45	80Y	113.35	99Y	115.25	118Y	117.15
62X	133.50	81X	113.40	100X	115.30	119X	117.20
62Y	133.55	81Y	113.45	100Y	115.35	119Y	117.25
63X	133.60	82X	113.50	101X	115.40	120X	117.30
63Y	133.65	82Y	113.55	101Y	115.45	120Y	117.35
64X	133.70	83X	113.60	102X	115.50	121X	117.40
64Y	133.75	83Y	113.65	102Y	115.55	121Y	117.45
65X	133.80	84X	113.70	103X	115.60	122X	117.50
65Y	133.85	84Y	113.75	103Y	115.65	122Y	117.55
66X	133.90	85X	113.80	104X	115.70	123X	117.60
66Y	133.95	85Y	113.85	104Y	115.75	123Y	117.65
67X	134.00	86X	113.90	105X	115.80	124X	117.70
67Y	134.05	86Y	113.95	105Y	115.85	124Y	117.75
68X	134.10	87X	114.00	106X	115.90	125X	117.80
68Y	134.15	87Y	114.05	106Y	115.95	125Y	117.85
69X	134.20	88X	114.10	107X	116.00	126X	117.90
69Y	134.25	88Y	114.15	107Y	116.05	126Y	117.95
70X	112.30	89X	114.20	108X	116.10		
70Y	112.35	89Y	114.25	108Y	116.15		
71X	112.40	90X	114.30	109X	116.20		
71Y	112.45	90Y	114.35	109Y	116.25		
72X	112.50	91X	114.40	110X	116.30		
72Y	112.55	91Y	114.45	110Y	116.35		
73X	112.60	92X	114.50	111X	116.40		
73Y	112.65	92Y	114.55	111Y	116.45		
74X	112.70	93X	114.60	112X	116.50		
74Y	112.75	93Y	114.65	112Y	116.55		
75X	112.80	94X	114.70	113X	116.60		
75Y	112.85	94Y	114.75	113Y	116.65		
76X	112.90	95X	114.80	114X	116.70		
76Y	112.95	95Y	114.85	114Y	116.75		

③ **COMM/NAV/WEATHER REMARKS:** These remarks consist of pertinent information affecting the current status of communications, NAVAIDS and weather.

Appendix 1

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TERMS/LANDING MINIMA DATA

IFR LANDING MINIMA

The United States Standard for Terminal Instrument Procedures (TERPS) is the approved criteria for formulating instrument approach procedures. Landing minima are established for six aircraft approach categories (ABCDE and COPTER). In the absence of COPTER MINIMA, helicopters may use the CAT A minimums of other procedures.

LANDING MINIMA FORMAT

In this example airport elevation is 1179, and runway touchdown zone elevation is 1152.

DA	Visibility (RVR 100's of feet)	Aircraft Approach Category HAT/HATh		
CATEGORY	A	B	C	D
S-ILS 27	1352/24		200 (200-½)	
S-LOC 27	1440/24	288 (300-½)		1440/50 288 (300-1)
CIRCLING	1540-1 361 (400-1)	1640-1 461 (500-1)	1640-1½ 461 (500-1½)	1740-2 561 (600-2)
MDA	HAA	Visibility in Statute Miles		

Straight-in ILS to Runway 27

Straight-in with Glide Slope Inoperative or not used to Runway 27

All weather minimums in parentheses not applicable to Civil Pilots. Military Pilots refer to appropriate regulations.

COPTER MINIMA ONLY

CATEGORY	COPTER		
H-176°	680-½	363 (400-½)	

Copter Approach Direction

Height of MDA/DA Above Landing Area (HAL)

No circling minimums are provided

RNAV (GPS) MINIMA EXAMPLE

CATEGORY	A	B	C	D
LPV DA	1540/24 258 (300-½)			
LNAV/VNAV DA	1600/24	318 (400-½)		1600/40 318 (400-¾)
LNAV MDA	1840/24	558 (600-½)	1840/50 558 (600-1)	1840/60 558 (600-1 ¼)
CIRCLING	1840-1	545 (600-1)	1840-1½ 545 (600-1½)	1860-2 565 (600-2)

NOTE: The **W** symbol indicates outages of the WAAS vertical guidance may occur daily at this location due to initial system limitations. WAAS NOTAMS for vertical outages are not provided for this approach. Use LNAV minima for flight planning at these locations, whether as a destination or alternate. For flight operations at these locations, when the WAAS avionics indicate that LNAV/VNAV or LPV service is available, then vertical guidance may be used to complete the approach using the displayed level of service. Should an outage occur during the procedure, reversion to LNAV minima may be required. As the WAAS coverage is expanded, the **W** will be removed.

RNAV minimums are dependent on navigation equipment capability, as stated in the applicable AFM, AFMS, or other FAA approved document. See AIM paragraph 5-4-5, AC 90-105 and AC 90-107 for detailed requirements for each line of minima.

AIRCRAFT APPROACH CATEGORIES

Aircraft approach category indicates a grouping of aircraft based on a speed of VREF, if specified, or if VREF not specified, 1.3 VSO at the maximum certificated landing weight. VREF, VSO, and the maximum certificated landing weight are those values as established for the aircraft by the certification authority of the country of registry. Helicopters are Category A aircraft. An aircraft shall fit in only one category. However, if it is necessary to operate at a speed in excess of the upper limit of the speed range for an aircraft's category, the minimums for the category for that speed shall be used. For example, an airplane which fits into Category B, but is circling to land at a speed of 145 knots, shall use the approach Category D minimums. As an additional example, a Category A airplane (or helicopter) which is operating at 130 knots on a straight-in approach shall use the approach Category C minimums. See following category limits:

MANEUVERING TABLE

Approach Category	A	B	C	D	E
Speed (Knots)	0-90	91-120	121-140	141-165	Abv 165

TERMS/LANDING MINIMA DATA

LEGEND 20.—Instrument Approach Procedures Explanation of Terms.

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TERMS/LANDING MINIMA DATA

CIRCLING APPROACH OBSTACLE PROTECTED AIRSPACE

The circling MDA provides vertical clearance from obstacles when conducting a circle-to-land maneuver within the obstacle protected area. Circling approach obstacle protected areas extend laterally and longitudinally from the centerlines and ends of all runways at an airport by the distances shown in the following tables. The areas are technically defined by the tangential connection of arcs drawn at the radius distance shown from each runway end.

STANDARD CIRCLING APPROACH MANEUVERING RADIUS

Circling approach protected areas developed prior to late 2012 used the radius distances shown in the following table, expressed in nautical miles (NM), dependent on aircraft approach category. The approaches using standard circling approach areas can be identified by the absence of the **C** symbol on the circling line of minima.

Circling MDA in feet MSL	Approach Category and Circling Radius (NM)				
	CAT A	CAT B	CAT C	CAT D	CAT E
All Altitudes	1.3	1.5	1.7	2.3	4.5

C EXPANDED CIRCLING APPROACH MANEUVERING AIRSPACE RADIUS

Circling approach protected areas developed after late 2012 use the radius distance shown in the following table, expressed in nautical miles (NM), dependent on aircraft approach category, and the altitude of the circling MDA, which accounts for true airspeed increase with altitude. The approaches using expanded circling approach areas can be identified by the presence of the **C** symbol on the circling line of minima.

Circling MDA in feet MSL	Approach Category and Circling Radius (NM)				
	CAT A	CAT B	CAT C	CAT D	CAT E
1000 or less	1.3	1.7	2.7	3.6	4.5
1001-3000	1.3	1.8	2.8	3.7	4.6
3001-5000	1.3	1.8	2.9	3.8	4.8
5001-7000	1.3	1.9	3.0	4.0	5.0
7001-9000	1.4	2.0	3.2	4.2	5.3
9001 and above	1.4	2.1	3.3	4.4	5.5

Comparable Values of RVR and Visibility

The following table shall be used for converting RVR to ground or flight visibility. For converting RVR values that fall between listed values, use the next higher RVR value; do not interpolate. For example, when converting 1800 RVR, use 2400 RVR with the resultant visibility of ½ mile.

RVR (feet)	Visibility (statute miles)	RVR (feet)	Visibility (statute miles)
1600	¼	4500	¾
2400	½	5000	1
3200	⅝	6000	1¼
4000	¾		

RADAR MINIMA

	RWY	GS/TCH/RPI	CAT	DA/ MDA-VIS	HAT/ HATH/ HAA	CEIL-VIS	CAT	DA/ MDA-VIS	HAT/ HATH/ HAA	CEIL-VIS
PAR	10	2.5°/42/1000	ABCDE	195/16	100	(100-¼)				
	28	2.5°/48/1068	ABCDE	187/16	100	(100-¼)				
ASR	10		ABC	560/40	463	(500-¾)	DE	560/50	463	(500-1)
	28		AB	600/50	513	(600-1)	CDE	600/60	513	(600-1¼)
CIR	10		AB	560-1¼	463	(500-1¼)	CDE	560-1½	463	(500-1½)
	28		AB	600-1¼	503	(600-1¼)	CDE	600-1½	503	(600-1½)

Visibility in Statute Miles

All minimums in parentheses not applicable to Civil Pilots. Military Pilots refer to appropriate regulations.

Radar Minima:

1. Minima shown are the lowest permitted by established criteria. Pilots should consult applicable directives for their category of aircraft.
2. The circling MDA and weather minima to be used are those for the runway to which the final approach is flown- not the landing runway. In the above RADAR MINIMA example, a category C aircraft flying a radar approach to runway 10, circling to land on runway 28, must use an MDA of 560 feet with weather minima of 500-1½.

NOTE: Military RADAR MINIMA may be shown with communications symbology that indicates emergency frequency monitoring capability by the radar facility as follows:

(E) VHF and UHF emergency frequencies monitored

(V) VHF emergency frequency (121.5) monitored

(U) UHF emergency frequency (243.0) monitored

Additionally, unmonitored frequencies which are available on request from the controlling agency may be annotated with an "x".

▲ Alternate Minimums not standard. Civil users refer to tabulation. USA/USN/USAF pilots refer to appropriate regulations.

▲ NA Alternate minimums are Not Authorized due to unmonitored facility or absence of weather reporting service.

▼ Takeoff Minimums not standard and/or Departure Procedures are published. Refer to tabulation.

TERMS/LANDING MINIMA DATA

LEGEND 21.—Instrument Approach Procedures Explanation of Terms.

GENERAL INFO

GENERAL INFORMATION

This publication is issued every 56 days and includes Standard Instrument Approach Procedures (SIAPs), Standard Instrument Departures (SIDs), Standard Terminal Arrivals (STARs), IFR Takeoff Minimums and (Obstacle) Departure Procedures (ODPs), IFR Alternate Minimums, and Radar Instrument Approach Minimums for use by civil and military aviation. The organization responsible for SIAPs, Radar Minimums, SIDs, STARs and graphic ODPs is identified in parentheses in the top margin of the procedure; e.g., (FAA), (FAA-O), (USA), (USAF), (USN). SIAPs with the (FAA) and (FAA-O) designation are regulated under 14 CFR, Part 97. SIAPs with the (FAA-O) designation have been developed under Other Transaction Agreement (OTA) by private providers and have been certified by the FAA. See 14 CFR, Part 91.175 (a) and the AIM for further details. 14 CFR, Part 91.175 (g) and the Special Notices section of the Airport/Facility Directory contains information on civil operations at military airports.

STANDARD TERMINAL ARRIVALS AND DEPARTURE PROCEDURES

The use of the associated codified STAR/DP and transition identifiers are requested of users when filing flight plans via teletype and are required for users filing flight plans via computer interface. It must be noted that when filing a STAR/DP with a transition, the first three coded characters of the STAR and the last three coded characters of the DP are replaced by the transition code. Examples: ACTON SIX ARRIVAL, file (AQN.AQN6); ACTON SIX ARRIVAL, EDNAS TRANSITION, file (EDNAS.AQN6). FREEHOLD THREE DEPARTURE, file (FREQ3.RBV), FREEHOLD THREE DEPARTURE, ELWOOD CITY TRANSITION, file (FREQ3.EWC).

RNAV DP and STAR. Effective March 15, 2007, these procedures, formerly identified as Type-A and Type-B, will be designated as RNAV 1 in accordance with amended Advisory Circular (AC) and ICAO terminology.

Refer to AC 90-100A U.S. TERMINAL AND EN ROUTE AREA NAVIGATION (RNAV) OPERATIONS and the Aeronautical Information Manual for additional guidance regarding these procedures.

Standard RNAV 1 Procedure Chart Notes

NOTE: RNAV 1

NOTE: DME/DME/IRU or GPS required

Some procedures may require use of GPS and will be identified by a "GPS required" note.

RNAV 1 Procedure Characteristics and Operations

1. Require use of an RNAV system with DME/DME/IRU, and/or GPS inputs.
2. Require use of a CDI, flight director, and/or autopilot, in lateral navigation mode, for flight guidance while operating on RNAV paths (track, course, or direct leg). Other methods providing an equivalent level of performance may be acceptable.
3. RNAV paths may start as low as 500 feet above airport elevation.

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ABBREVIATIONS

AAUP.....	Attention All Users Page	HAA.....	Height above Airport
ADF.....	Automatic Direction Finder	HAL.....	Height above Landing
AFIS.....	Automatic Flight Information Service	HAT.....	Height above Touchdown
ALS.....	Approach Light System	HATh.....	Height Above Threshold
ALSF.....	Approach Light System with Sequenced Flashing Lights	HGS.....	Head-up Guidance System
AP.....	Autopilot System	HIRL.....	High Intensity Runway Lights
APCH.....	Approach	HUD.....	Head-up Display
APP CON.....	Approach Control	IAF.....	Initial Approach Fix
ARR.....	Arrival	ICAO.....	International Civil Aviation Organization
ASOS.....	Automated Surface Observing System	IF.....	Intermediate Fix
ASR/PAR.....	Published Radar Minimums at this Airport	IM.....	Inner Marker
ATIS.....	Automatic Terminal Information Service	INT.....	Intersection
AUNICOM.....	Automated UNICOM	LAAS.....	Local Area Augmentation System
AWOS.....	Automated Weather Observing System	LDA.....	Localizer Type Directional Aid
AZ.....	Azimuth	Ldg.....	Landing
BC.....	Back Course	LRL.....	Low Intensity Runway Lights
BND.....	Bound	LNAV.....	Lateral Navigation
C.....	Circling	LOC.....	Localizer
CAT.....	Category	LP.....	Localizer Performance
CCW.....	Counter Clockwise	LPV.....	Localizer Performance with Vertical Guidance
CDI.....	Course Deviation Indicator	LR.....	Lead Radial. Provides at least 2 NM (Copter 1 NM) of lead to assist in turning onto the intermediate/final course.
Chan.....	Channel	MAA.....	Maximum Authorized Altitude
CIR.....	Circling	MALS.....	Medium Intensity Approach Light System
CLNC DEL.....	Clearance Delivery	MALSR.....	Medium Intensity Approach Light System with RAIL
CNF.....	Computer Navigation Fix	MAP.....	Missed Approach Point
CTAF.....	Common Traffic Advisory Frequency	MDA.....	Minimum Descent Altitude
CW.....	Clockwise	MIRL.....	Medium Intensity Runway Lights
DA.....	Decision Altitude	MLS.....	Microwave Landing System
DER.....	Departure End of Runway	MM.....	Middle Marker
DH.....	Decision Height	MRA.....	Minimum Reception Altitude
DME.....	Distance Measuring Equipment	N/A.....	Not Applicable
DTHR.....	Displaced Threshold	NA.....	Not Authorized
ELEV.....	Elevation	NDB.....	Non-directional Radio Beacon
EMAS.....	Engineered Material Arresting System	NFD.....	National Flight Database
FAF.....	Final Approach Fix	NM.....	Nautical Mile
FD.....	Flight Director System	NoPT.....	No Procedure Turn Required (Procedure Turn shall not be executed without ATC clearance)
FM.....	Fan Marker	ODALS.....	Omnidirectional Approach Light System
FMS.....	Flight Management System	ODP.....	Obstacle Departure Procedure
GCO.....	Ground Communications Outlet	OM.....	Outer Marker
GLS.....	Ground Based Augmentation System Landing System	PRM.....	Precision Runway Monitor
GPI.....	Ground Point of Interception		
GPS.....	Global Positioning System		
GS.....	Glide Slope		

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LEGEND 23.—Abbreviations.

Appendix 1

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GENERAL INFO

ABBREVIATIONS

R.....	Radial
RA.....	Radio Altimeter setting height
RAIL.....	Runway Alignment Indicator Lights
RCLS.....	Runway Centerline Light System
REIL.....	Runway End Identifier Lights
RF.....	Radius-to-Fix
RLLS.....	Runway Lead-in Light System
RNAV.....	Area Navigation
RNP.....	Required Navigation Performance
RPI.....	Runway Point of Intercept(ion)
RRL.....	Runway Remaining Lights
Rwy.....	Runway
RVR.....	Runway Visual Range
S.....	Straight-in
SALS.....	Short Approach Light System
SSALR.....	Simplified Short Approach Light System with RAIL
SDF.....	Simplified Directional Facility
SM.....	Statute Mile
SOIA.....	Simultaneous Offset Instrument Approach
TAA.....	Terminal Arrival Area
TAC.....	TACAN
TCH.....	Threshold Crossing Height (height in feet Above Ground level)
TDZ.....	Touchdown Zone
TDZE.....	Touchdown Zone Elevation
TDZ/CL.....	Touchdown Zone and Runway Centerline Lighting
TDZL.....	Touchdown Zone Lights
THR.....	Threshold
THRE.....	Threshold Elevation
TODA.....	Takeoff Distance Available
TORA.....	Takeoff Run Available
TR.....	Track
VASI.....	Visual Approach Slope Indicator
VCOA.....	Visual Climb Over Airport
VDP.....	Visual Descent Point
VGSI.....	Visual Glide Slope Indicator
VNAV.....	Vertical Navigation
WAAS.....	Wide Area Augmentation System
WP/WPT.....	Waypoint (RNAV)

GENERAL INFO

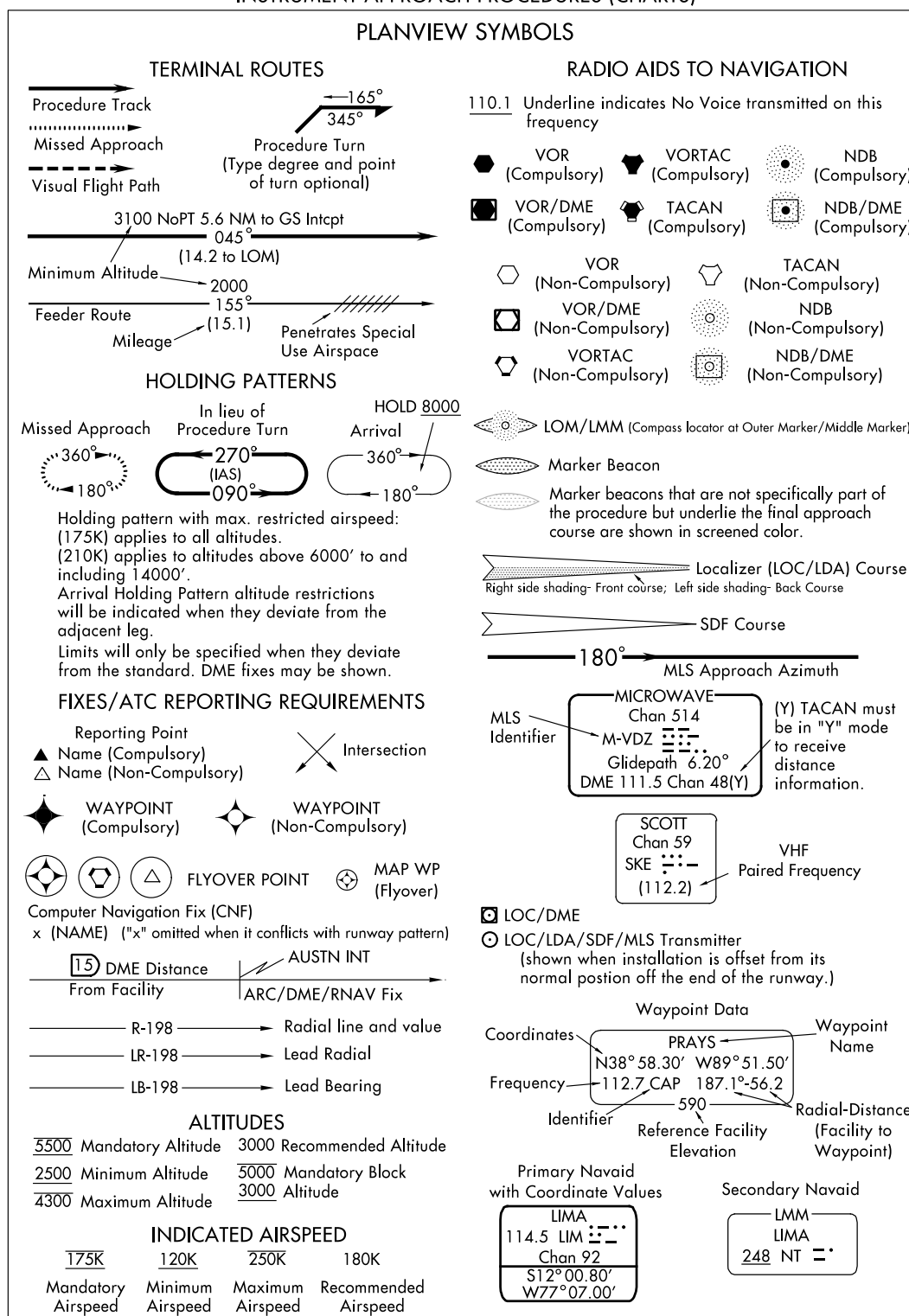
14149

LEGEND 23A.—Abbreviations.

11349

LEGEND

INSTRUMENT APPROACH PROCEDURES (CHARTS)

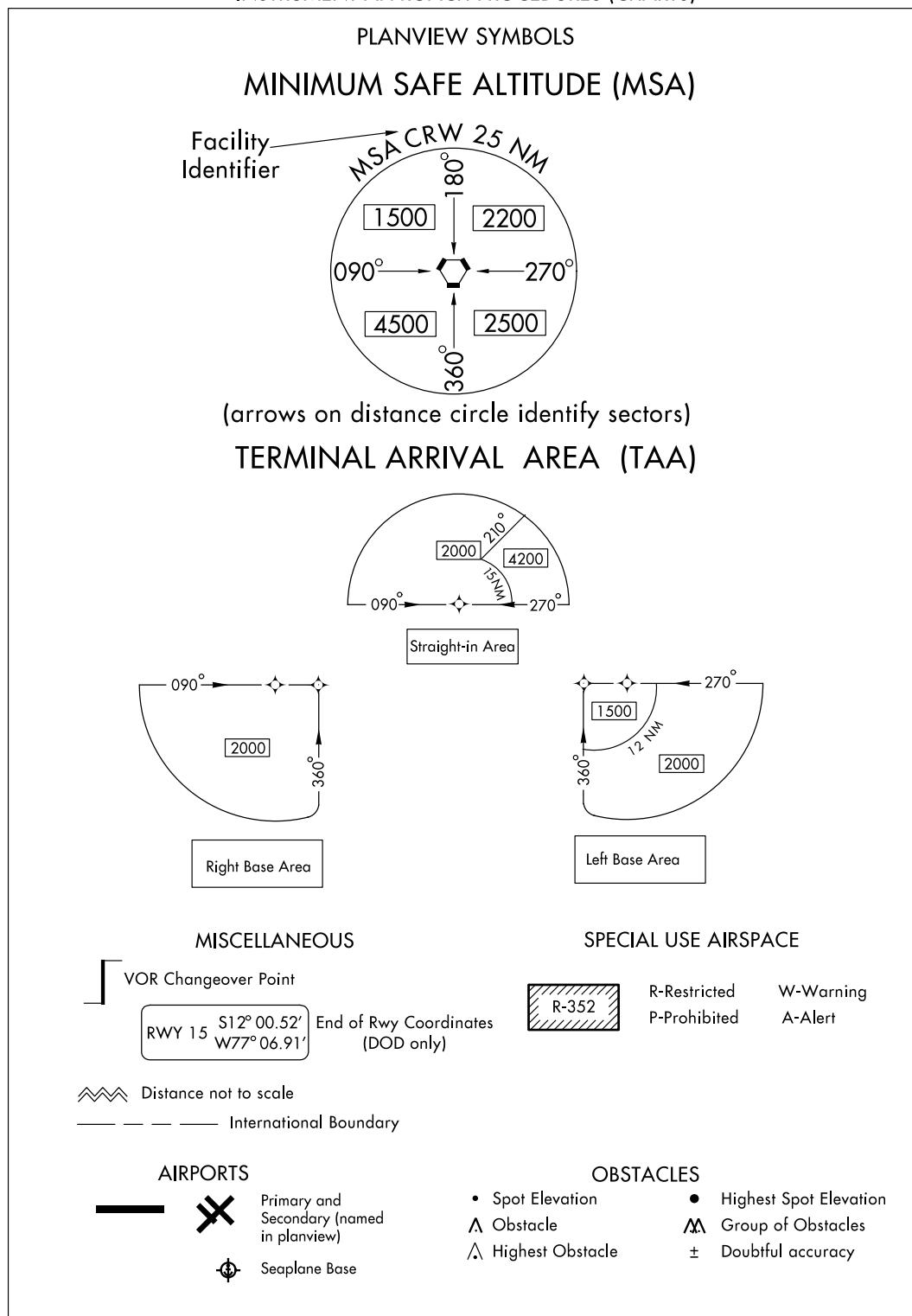


LEGEND

LEGEND 24.—Instrument Approach Procedures (Symbols).

LEGEND

INSTRUMENT APPROACH PROCEDURES (CHARTS)



LEGEND

LEGEND 25.—Instrument Approach Procedures (Symbols).

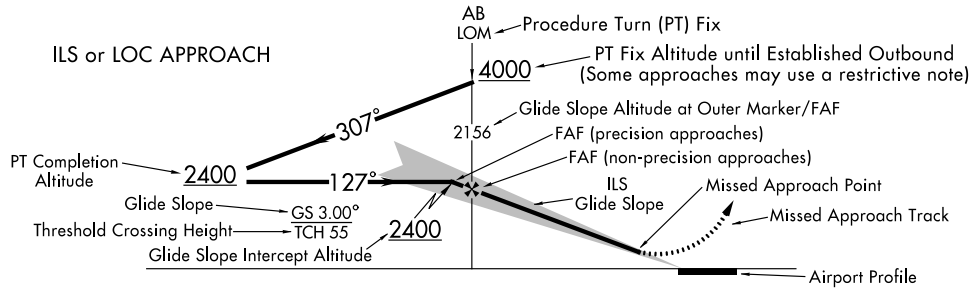
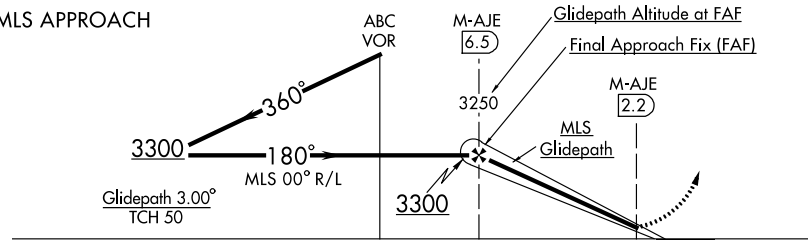
13346

LEGEND**INSTRUMENT APPROACH PROCEDURES (CHARTS)****PROFILE VIEW**

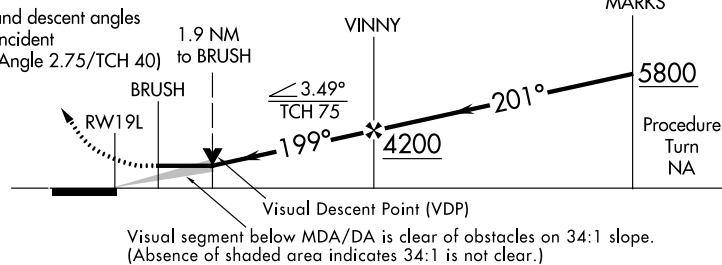
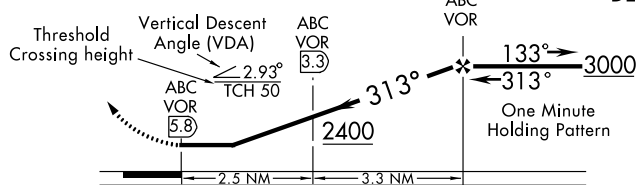
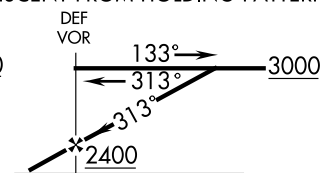
Two different methods are used for vertical guidance:

a. "GS" indicates an electronic glide slope or barometric vertical guidance is present. In the case of an Instrument Landing System (ILS) and Wide Area Augmentation System (WAAS) LPV approach procedures, an electronic signal provides vertical guidance. Barometric vertical guidance is provided for RNP and LNAV/VNAV instrument approach procedures. All ILS, LPV, RNP, and LNAV/VNAV will be in this format $\text{GS } 3.00^\circ$, located in the lower left or right corner.

b. Other charts without electronic or barometric vertical guidance will be in this format $\text{TCH } 55^\circ$, indicating a non-precision vertical descent angle to assist in preventing controlled flight into terrain. On Civil (FAA) procedures, this information is placed above or below the procedure track following the fix it is based on.

**MLS APPROACH****RNAV APPROACH**

VGSI and descent angles not coincident (VGSI Angle 2.75/TCH 40)

**NON PRECISION****DESCENT FROM HOLDING PATTERN****ALTITUDES**

5500	Mandatory Altitude	3000	Recommended Altitude
2500	Minimum Altitude	5000	Mandatory Block Altitude
4300	Maximum Altitude	3000	Altitude

PROFILE SYMBOLS

	Glide Slope/Glide Path Intercept Altitude and final approach fix for vertically guided approach procedures.		Visual Flight Path
	Visual Descent Point (VDP)		Note: Facilities and waypoints are depicted as a solid vertical line while fixes and intersections are depicted as a dashed vertical line.

LEGEND

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LEGEND 26.—Instrument Approach Procedures (Profile).

Appendix 1

CLIMB/DESCENT TABLE 10042

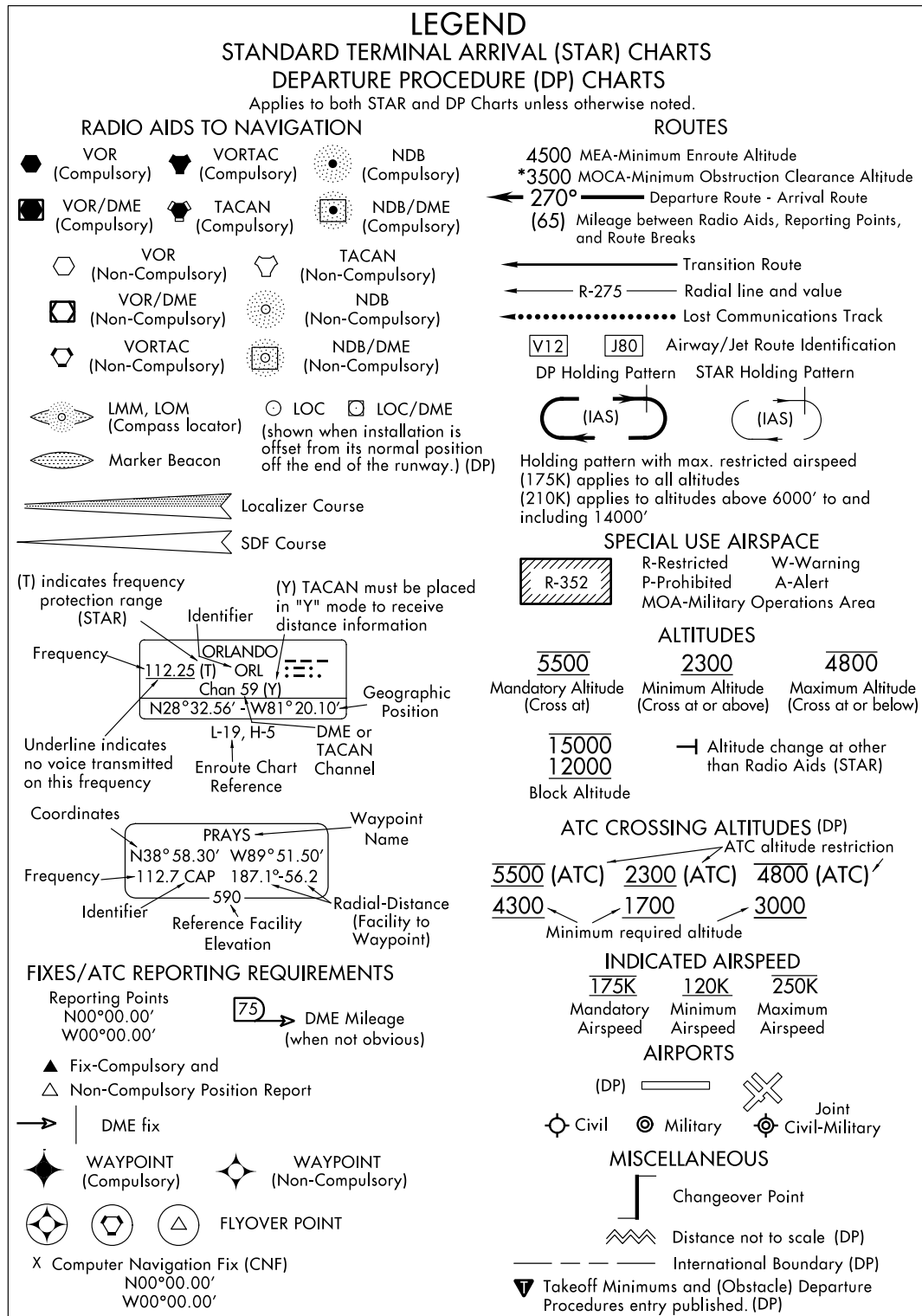
INSTRUMENT TAKEOFF OR APPROACH PROCEDURE CHARTS RATE OF CLIMB/DESCENT TABLE (ft. per min)												
A rate of climb/descent table is provided for use in planning and executing climbs or descents under known or approximate ground speed conditions. It will be especially useful for approaches when the localizer only is used for course guidance. A best speed, power, altitude combination can be programmed which will result in a stable glide rate and altitude favorable for executing a landing if minimums exist upon breakout. Care should always be exercised so that minimum descent altitude and missed approach point are not exceeded.												
CLIMB/ DESCENT ANGLE (degrees and tenths)	ft/NM	GROUND SPEED (knots)										
		60	90	120	150	180	210	240	270	300	330	360
2.0	210	210	320	425	530	635	743	850	955	1060	1165	1275
2.5	265	265	400	530	665	795	930	1060	1195	1325	1460	1590
VERTICAL PATH ANGLE	2.7	287	287	430	574	717	860	1003	1147	1290	1433	1576
	2.8	297	297	446	595	743	892	1041	1189	1338	1486	1635
	2.9	308	308	462	616	770	924	1078	1232	1386	1539	1693
	3.0	318	318	478	637	797	956	1115	1274	1433	1593	1752
	3.1	329	329	494	659	823	988	1152	1317	1481	1646	1810
	3.2	340	340	510	680	850	1020	1189	1359	1529	1699	1869
	3.3	350	350	526	701	876	1052	1227	1402	1577	1752	1927
	3.4	361	361	542	722	903	1083	1264	1444	1625	1805	1986
3.5	370	370	555	745	930	1115	1300	1485	1670	1860	2045	2230
4.0	425	425	640	850	1065	1275	1490	1700	1915	2125	2340	2550
4.5	480	480	715	955	1195	1435	1675	1915	2150	2390	2630	2870
5.0	530	530	795	1065	1330	1595	1860	2125	2390	2660	2925	3190
5.5	585	585	880	1170	1465	1755	2050	2340	2635	2925	3220	3510
6.0	640	640	960	1275	1595	1915	2235	2555	2875	3195	3510	3830
6.5	690	690	1040	1385	1730	2075	2425	2770	3115	3460	3805	4155
7.0	745	745	1120	1490	1865	2240	2610	2985	3355	3730	4105	4475
7.5	800	800	1200	1600	2000	2400	2800	3200	3600	4000	4400	4800
8.0	855	855	1280	1710	2135	2560	2990	3415	3845	4270	4695	5125
8.5	910	910	1360	1815	2270	2725	3180	3630	4085	4540	4995	5450
9.0	960	960	1445	1925	2405	2885	3370	3850	4330	4810	5295	5775
9.5	1015	1015	1525	2035	2540	3050	3560	4065	4575	5085	5590	6100
10.0	1070	1070	1605	2145	2680	3215	3750	4285	4820	5355	5890	6430

CLIMB/DESCENT TABLE 10042

LEGEND 27.—Instrument Takeoff or Approach Procedure Charts, Rate-of-Climb/Descent Table.

12320

LEGEND

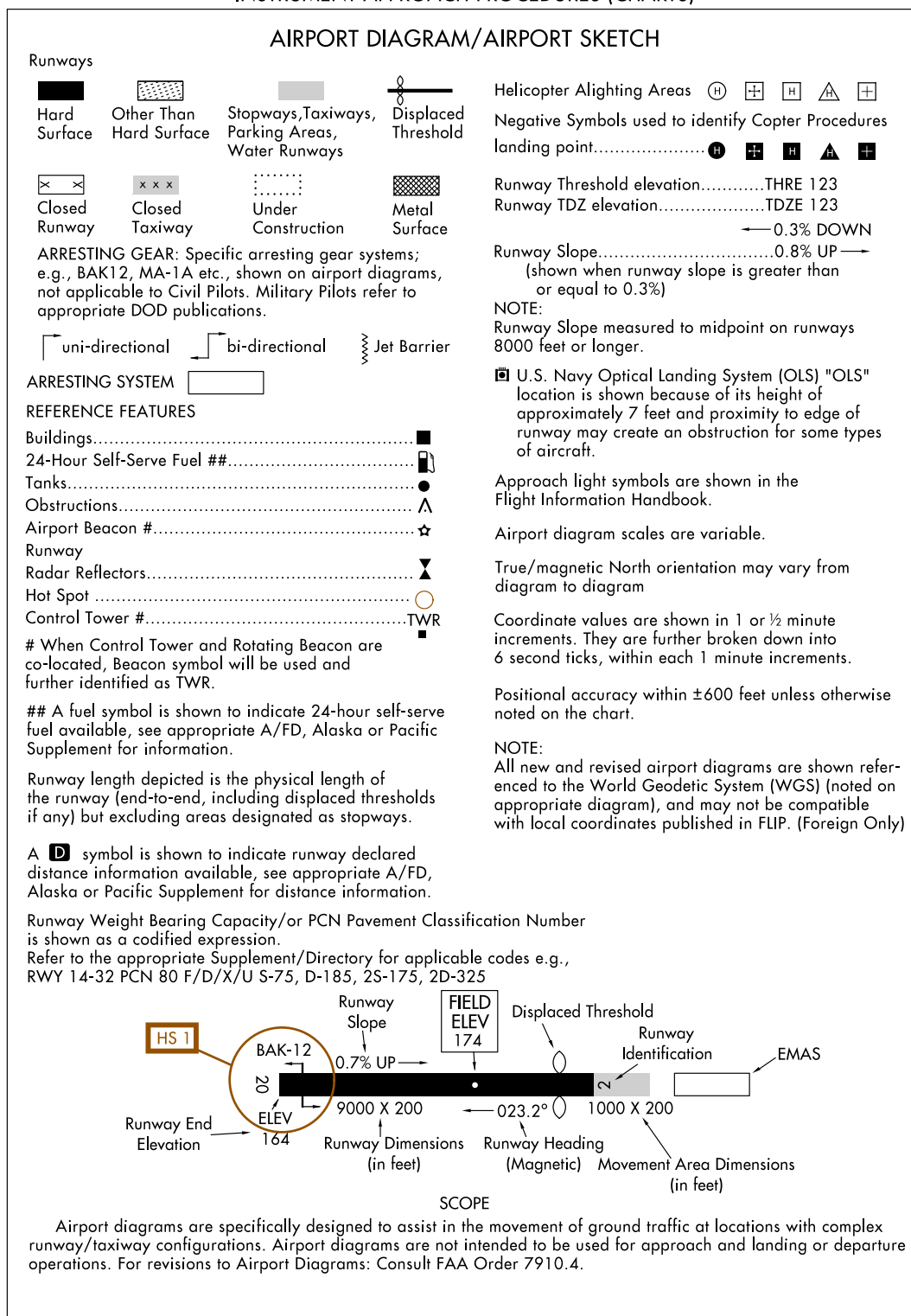


LEGEND

LEGEND 28.—Standard Arrival/Departure Charts.

LEGEND

INSTRUMENT APPROACH PROCEDURES (CHARTS)



LEGEND

LEGEND 29.—Airport Diagram.

09239

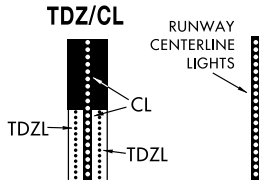
LEGEND

INSTRUMENT APPROACH PROCEDURES (CHARTS)
APPROACH LIGHTING SYSTEM - UNITED STATES

Approach lighting and visual glide slope systems are indicated on the airport sketch by an identifier, e.g., (A₂), (V), etc.

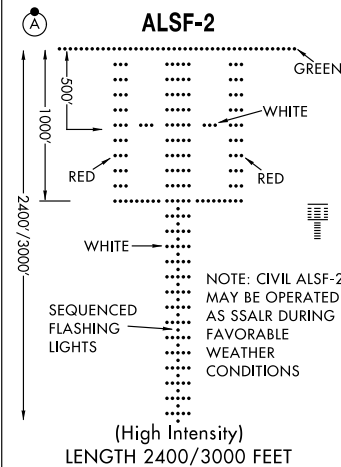
A dot "•" portrayed with approach lighting letter identifier indicates sequenced flashing lights (F) installed with the approach lighting system e.g., (A₁). Negative symbology, e.g., (A₁), (V) indicates Pilot Controlled Lighting (PCL).

**RUNWAY TOUCHDOWN ZONE
AND CENTERLINE
LIGHTING SYSTEMS**

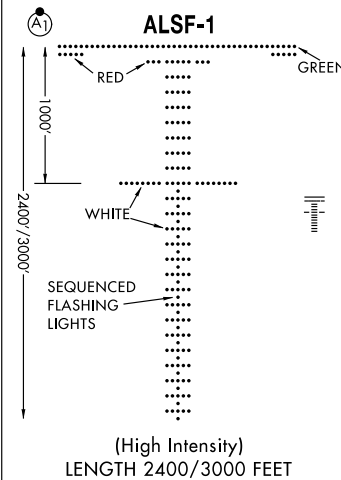


AVAILABILITY of TDZ/CL will be shown by
NOTE in SKETCH e.g. "TDZ/CL Rwy 15"

APPROACH LIGHTING SYSTEM



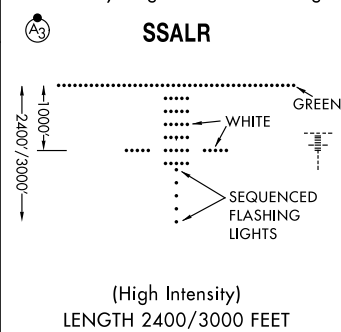
APPROACH LIGHTING SYSTEM



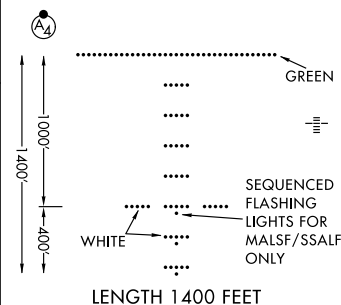
**SHORT APPROACH
LIGHTING SYSTEM**

SALS/SALSF
(High Intensity)
SAME AS INNER 1500' OF ALSF-1

**SIMPLIFIED SHORT
APPROACH LIGHTING SYSTEM**
with Runway Alignment Indicator Lights



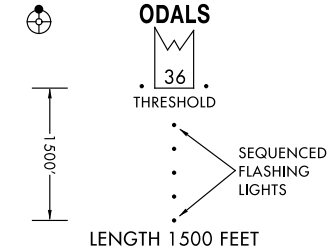
**MEDIUM INTENSITY (MALS and
MALSF) OR SIMPLIFIED SHORT
(SSALS and SSALF)
APPROACH LIGHTING SYSTEMS**



**MEDIUM INTENSITY
APPROACH LIGHTING SYSTEM**
with Runway Alignment Indicator Lights

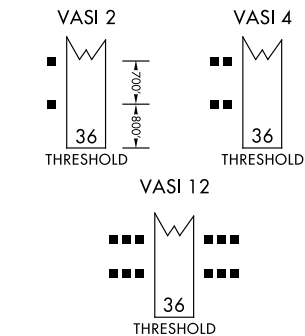
MALSR
SAME LIGHT CONFIGURATION
AS SSALR.

**OMNIDIRECTIONAL
APPROACH LIGHTING SYSTEM**



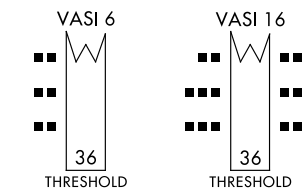
**VISUAL APPROACH
SLOPE INDICATOR**

VASI
VISUAL APPROACH SLOPE INDICATOR
WITH STANDARD THRESHOLD CLEARANCE
PROVIDED.
ALL LIGHTS WHITE — TOO HIGH
FAR LIGHTS RED — ON GLIDE SLOPE
NEAR LIGHTS WHITE
ALL LIGHTS RED — TOO LOW



**VISUAL APPROACH
SLOPE INDICATOR**

VASI
VISUAL APPROACH SLOPE INDICATOR
WITH A THRESHOLD CROSSING HEIGHT TO
ACCOMMODATE LONG BODIED OR JUMBO
AIRCRAFT.

**LEGEND**

LEGEND 30.—Approach Lighting Systems.

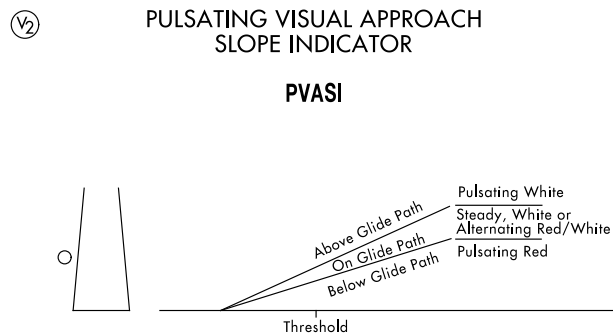
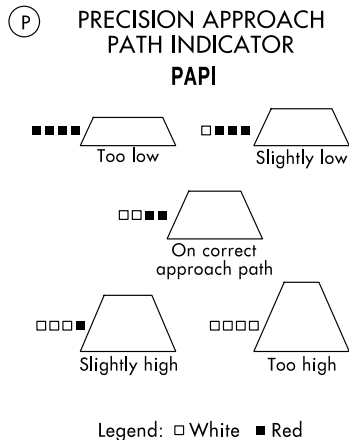
04330

LEGEND

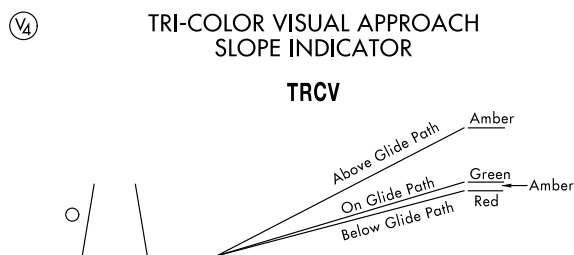
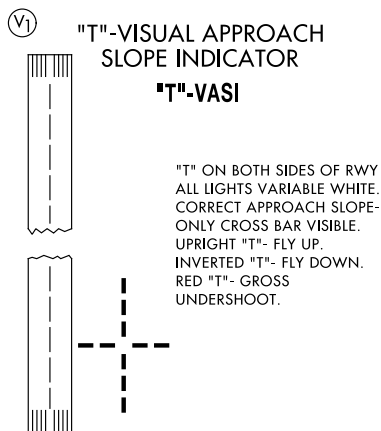
INSTRUMENT APPROACH PROCEDURES (CHARTS)
APPROACH LIGHTING SYSTEM - UNITED STATES

Approach lighting and visual glide slope systems are indicated on the airport sketch by an identifier, (A2), (V) etc.

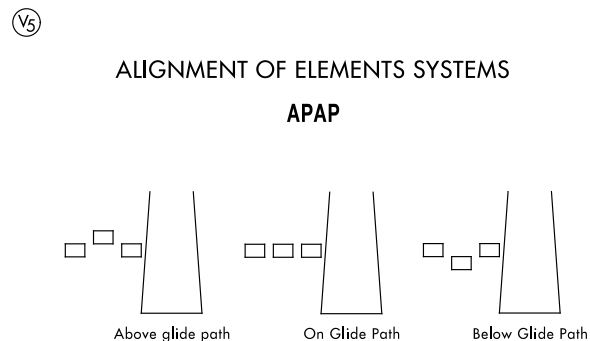
A dot "•" portrayed with approach lighting letter identifier indicates sequenced flashing lights (F) installed with the approach lighting system e.g., (A1). Negative symbology, e.g., (A1), (V) indicates Pilot Controlled Lighting (PCL).



CAUTION: When viewing the pulsating visual approach slope indicators in the pulsating white or pulsating red sectors, it is possible to mistake this lighting aid for another aircraft or a ground vehicle. Pilots should exercise caution when using this type of system.



CAUTION: When the aircraft descends from green to red, the pilot may see a dark amber color during the transition from green to red.



Painted panels which may be lighted at night.
 To use the system the pilot positions the aircraft
 so the elements are in alignment.

LEGEND

LEGEND 31.—Approach Lighting System.

13122

INOP COMPONENTS

INOPERATIVE COMPONENTS OR VISUAL AIDS TABLE

Landing minimums published on instrument approach procedure charts are based upon full operation of all components and visual aids associated with the particular instrument approach chart being used. Higher minimums are required with inoperative components or visual aids as indicated below. If more than one component is inoperative, each minimum is raised to the highest minimum required by any single component that is inoperative. ILS glide slope inoperative minimums are published on the instrument approach charts as localizer minimums. This table may be amended by notes on the approach chart. Such notes apply only to the particular approach category(ies) as stated. See legend page for description of components indicated below.

(1) ILS, MLS, PAR and RNAV (LPV line of minima)

Inoperative Component or Aid	Approach Category	Increase Visibility
ALSF 1 & 2, MALSR, & SSALR	ABCD	¼ mile

(2) ILS with visibility minimum of 1,800 RVR

ALSF 1 & 2, MALSR, & SSALR	ABCD	To 4000 RVR
TDZL RCLS	ABCD	To 2400 RVR*
RVR	ABCD	To ½ mile

*1800 RVR authorized with the use of FD or AP or HUD to DA.

(3) VOR, VOR/DME, TACAN, LOC, LOC/DME, LDA, LDA/DME, SDF, SDF/DME, GPS, ASR and RNAV (LNAV/VNAV, LP, LNAV lines of minima)

Inoperative Visual Aid	Approach Category	Increase Visibility
ALSF 1 & 2, MALSR, & SSALR	ABCD	½ mile
SSALS, MALS, & ODALS	ABC	¼ mile

(4) NDB

ALSF 1 & 2, MALSR, & SSALR	C	½ mile
MALS, SSALS, ODALS	ABD	¼ mile
	ABC	¼ mile

INOP COMPONENTS

LEGEND 32.—Inoperative Components or Visual Aids Table.

L-13
PANELS
ABCDE
1"=16 NM

UNITED STATES GOVERNMENT
FLIGHT INFORMATION PUBLICATION

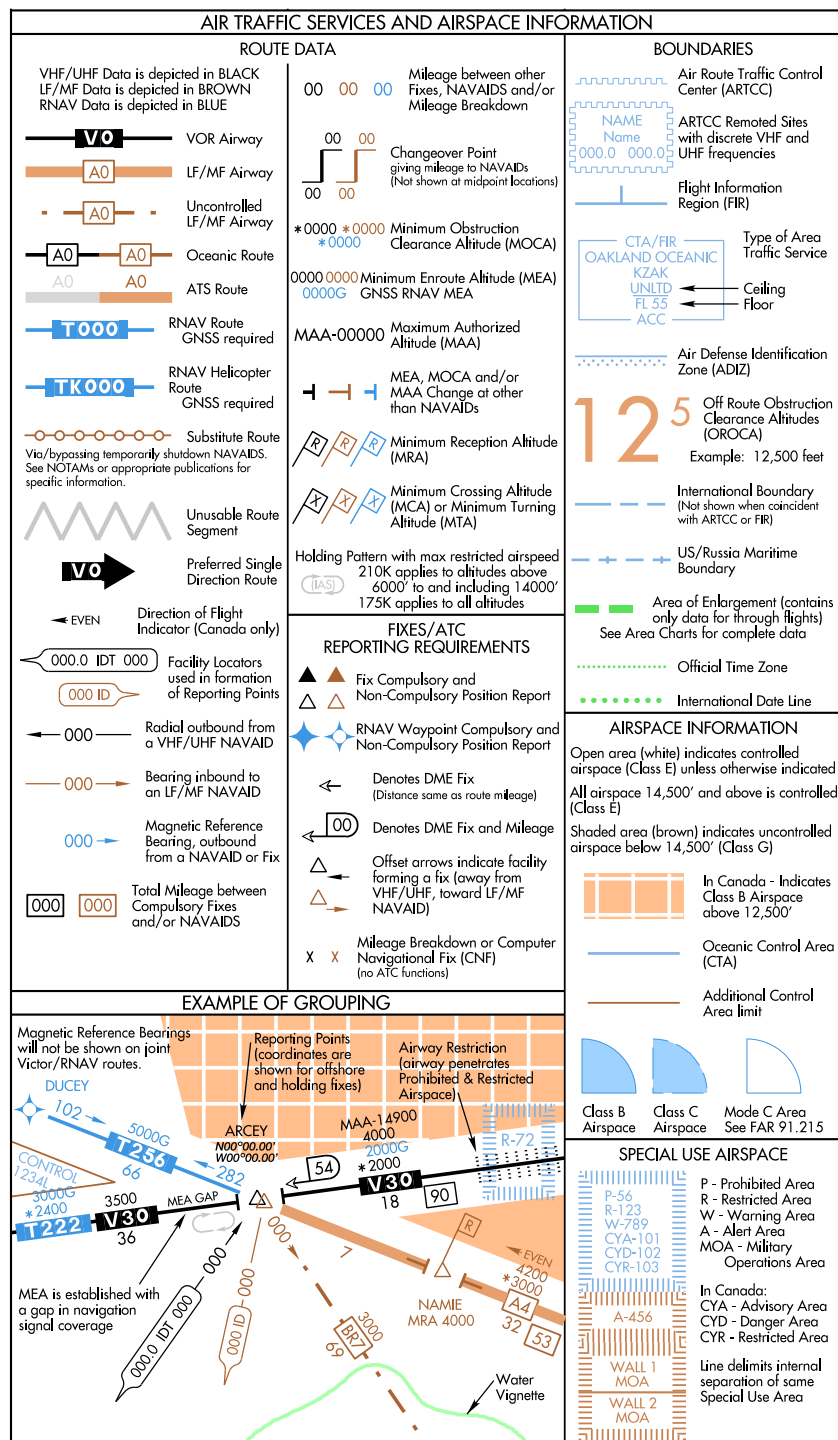
L-14
PANELS
FGHIJ
1"=14 NM

IFR ENROUTE LOW ALTITUDE - U.S.

For use up to but not including 18,000' MSL
HORIZONTAL DATUM: NORTH AMERICAN DATUM OF 1983

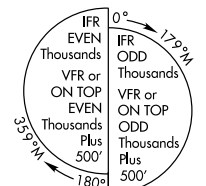
L E G E N D		
AIRPORTS		
Facilities in BLUE or GREEN have an approved Instrument Approach Procedure and/or RADAR MINIMA published in either the FAA Terminal Procedures Publications or the DoD FIPs. Those in BLUE have an Instrument Approach Procedure and/or RADAR MINIMA published at least in the High Altitude DoD FIPs. Facilities in BROWN do not have a published Instrument Approach Procedure or RADAR MINIMA.		
LAND Civil Civil - Military Military Heliport Heliport Heliport SEA Civil Civil Civil	1. A solid line box enclosing the airport name indicates FAR 93 Special Requirements—see Directory/Supplement 2. "NO SVFR" above the airport name indicates FAR 91 fixed-wing special VFR flight is prohibited 3. [C] or [D] following the airport name indicates Class C or Class D Airspace	4. Pvt - Private use 5. Associated city names for public airports are shown above or preceding the airport name. If airport name and city name are the same, only the airport name is shown. City names for military and private airports are not shown. The airport identifier in parentheses follows the airport name.
Lighting Capability: L Lighting available Q Pilot Controlled Lighting ★ Part-time or on request - No lighting available At private facilities - indicates no lighting information is available.		
NAVAIDS		
 VHF/UHF Data is depicted in BLACK LF/MF Data is depicted in BROWN COMPASS ROSE and/or NORTH ARROW Oriented to Magnetic North of NAVAID which may not be adjusted to the charted isogonic values. Smaller sizes are used in congested areas. Compass Locator Beacon	VOR VOR/DME TACAN VORTAC LF/MF Non-directional Radiobeacon or Marine Radiobeacon LF/MF Non-directional Radiobeacon /DME	Non Compulsory Reporting or Off Airway Compulsory Reporting
Flight Service Station (FSS), Remote Communications Outlet (RCO) or Automated Weather Observing Station (ASOS/AWOS) not associated with a charted NAVAID or airport		ILS Localizer Course with additional navigation function
COMMUNICATION BOXES		
 NAME (T) 000.0 IDT 000(Y) :::: VOR with TACAN compatible DME Underline indicates No Voice transmitted on this frequency. TACAN Channels are without voice but not underlined. Crosshatch indicates Shutdown status (T) Frequency protection usable range at 12,000' AGL - 25NM (Y) TACAN must be placed in "Y" mode to receive distance information NAME 000 IDT :::: (000.0) N00°00.00' W000°00.00' TACAN Channel paired with VHF Frequency in parenthesis. Automated Weather Broadcast Systems: ASOS/AWOS HIWAS Automated weather, when available, is broadcast on the associated NAVAID frequency. NAME ASOS 000.0 Stand Alone ASOS/AWOS	 000.0 NAME 000.0 IDT 000 :::: N00°00.00' W000°00.00' [NAME] ← FSS name Freq(s) positioned above thin line NAVAID box is remoted to the NAVAID site. Other freq(s) at the named FSS radio are available, however, altitude and terrain may determine their reception. Thin line NAVAID boxes without freq(s) and FSS radio name indicates no freq(s) available. 000.0 000.0 NAME 000.0 IDT 000 :::: N00°00.00' W000°00.00' Shadow NAVAID box indicates NAVAID and Flight Service Station (FSS) have same name 000.0 000.0 NAME IDT FSS name and identifier not associated with NAVAID NAME 000.0 Remote Communications Outlet (RCO). FSS radio name and remoted freq(s) are shown.	 Part-Time or On-Request NAME ★ 000 IDT 00(000.0) :::: LF/MF Non-directional Radiobeacon/DME VHF Freq paired with TACAN Channel SHADOW BOXES indicate Flight Service Stations (FSS). Frequencies 122.2, 255.4 and emergency 121.5 and 243.0 (Canada-121.5, 126.7 and 243.0) are available at many FSSs and are not shown. All other frequencies are shown. Certain FSSs provide Airport Advisory Service, see A/FD. Frequencies transmit and receive except those followed by R or T: R - Receive only T - Transmit only In Canada, a "D" after the frequency indicates a Dial-up Remote Communications Outlet.

LEGEND 33.—IFR En Route Low Altitude (U.S.)



LEGEND 34.—IFR En Route Low Altitude (U.S.)

Appendix 1

MILITARY TRAINING ROUTES (MTRs)	CRUISING ALTITUDES - U.S.	MISCELLANEOUS
MTRs 5 NM or less both sides of centerline IR-000 → VR-000 → MTRs greater than 5 NM either or both sides of centerline IR-000 → VR-000 → Arrow indicates direction of route See MTR tabs for altitude range information All IR and VR MTRs are shown except those VRs at or below 1500' AGL CAUTION: Inset charts do not depict MTRs	IFR within controlled airspace as assigned by ATC  VFR above 3000' AGL unless otherwise authorized by ATC IFR outside controlled airspace All courses are magnetic	ALTIMETER Altimeter setting change 4°E 2010 Isogonic Line and Value All Mileages are Nautical except as noted. All Radials and Bearings are magnetic except as noted. All Altitudes are MSL except as noted. All Time is Coordinated Universal Time (UTC), Days are local. ‡ During periods of Daylight Saving Time (DT), effective hours will be one hour earlier than shown. All states observe DT except Arizona and Hawaii. North American Datum of 1983 (NAD 83), for charting purposes is considered equivalent to World Geodetic System 1984 (WGS 84). FOR ADDITIONAL SYMBOL INFORMATION REFER TO THE CHART USER'S GUIDE

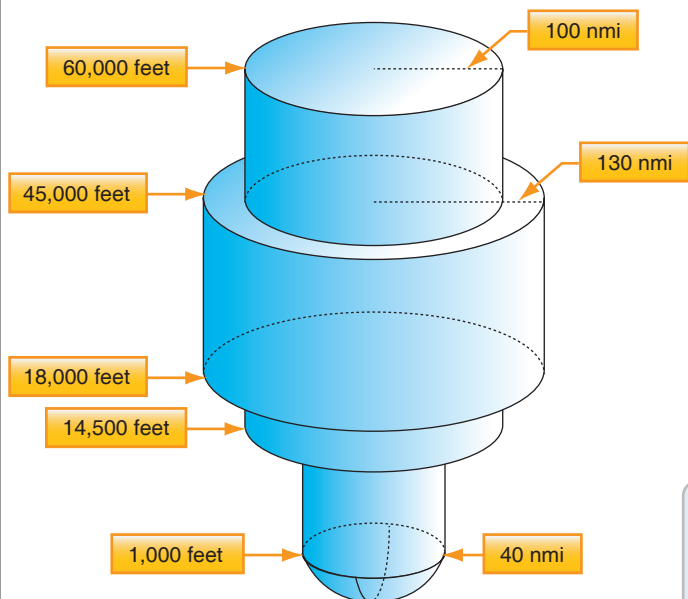
LEGEND 35.—IFR En Route Low Altitude (U.S.)

AIRCRAFT EQUIPMENT SUFFIXES	
SUFFIX	Aircraft Equipment Suffixes
	NO DME
/X	No transponder
/T	Transponder with no Mode C
/U	Transponder with Mode C
	DME
/D	No transponder
/B	Transponder with no Mode C
/A	Transponder with Mode C
	TACAN ONLY
/M	No transponder
/N	Transponder with no Mode C
/P	Transponder with Mode C
	AREA NAVIGATION (RNAV)
/Y	LORAN, VOR/DME, or INS with no transponder
/C	LORAN, VOR/DME, or INS, transponder with no Mode C
/I	LORAN, VOR/DME, or INS, transponder with Mode C
	ADVANCED RNAV WITH TRANSPONDER AND MODE C (if an aircraft is unable to operate with a transponder and/or Mode C, it will revert to the appropriate code listed above under Area Navigation.)
/E	Flight Management System (FMS) with en route, terminal, and approach capability. Equipment requirements are: (a) Dual FMS which meets the specifications of AC 25-15, Approval of Flight Management Systems in Transport Category Airplanes; AC 20-129, Airworthiness Approval of Vertical Navigation (VNAV) Systems for use in the U.S. NAS and Alaska; AC 20-130, Airworthiness Approval of Navigation or Flight Management Systems Integrating Multiple Navigation Sensors; or equivalent criteria as approved by Flight Standards. (b) A flight director and autopilot control system capable of following the lateral and vertical FMS flight path. (c) At least dual inertial reference units (IRU's). (d) A database containing the waypoints and speed/altitude constraints for the route and/or procedure to be flown that is automatically loaded into the FMS flight plan. (e) An electronic map. (U.S. and U.S. territories only unless otherwise authorized.)
/F	A single FMS with en route, terminal, and approach capability that meets the equipment requirements of /E, (a) through (d), above. (U.S. and U.S. territories only unless otherwise authorized.)
/G	Global Positioning System (GPS)/Global Navigation Satellite System (GNSS) equipped aircraft with en route and terminal capability.
/R	Required Navigational Performance (Denotes capability to operate in RNP designated airspace and routes)
/W	Reduced Vertical Separation Minima (RVSM)

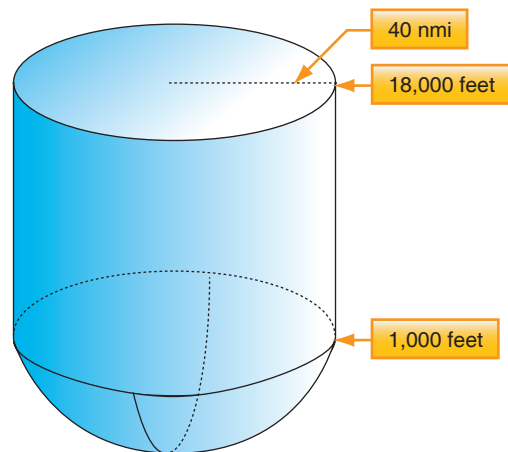
LEGEND 36.—Aircraft Equipment Suffixes.

AIR NAVIGATION RADIO AIDS

Standard High Altitude Service Volume

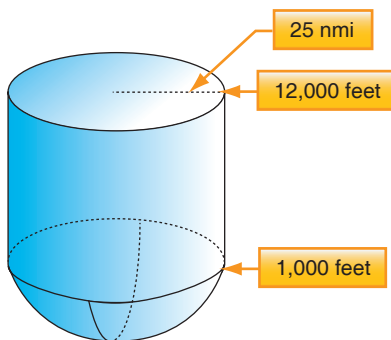


Standard Low Altitude Service Volume

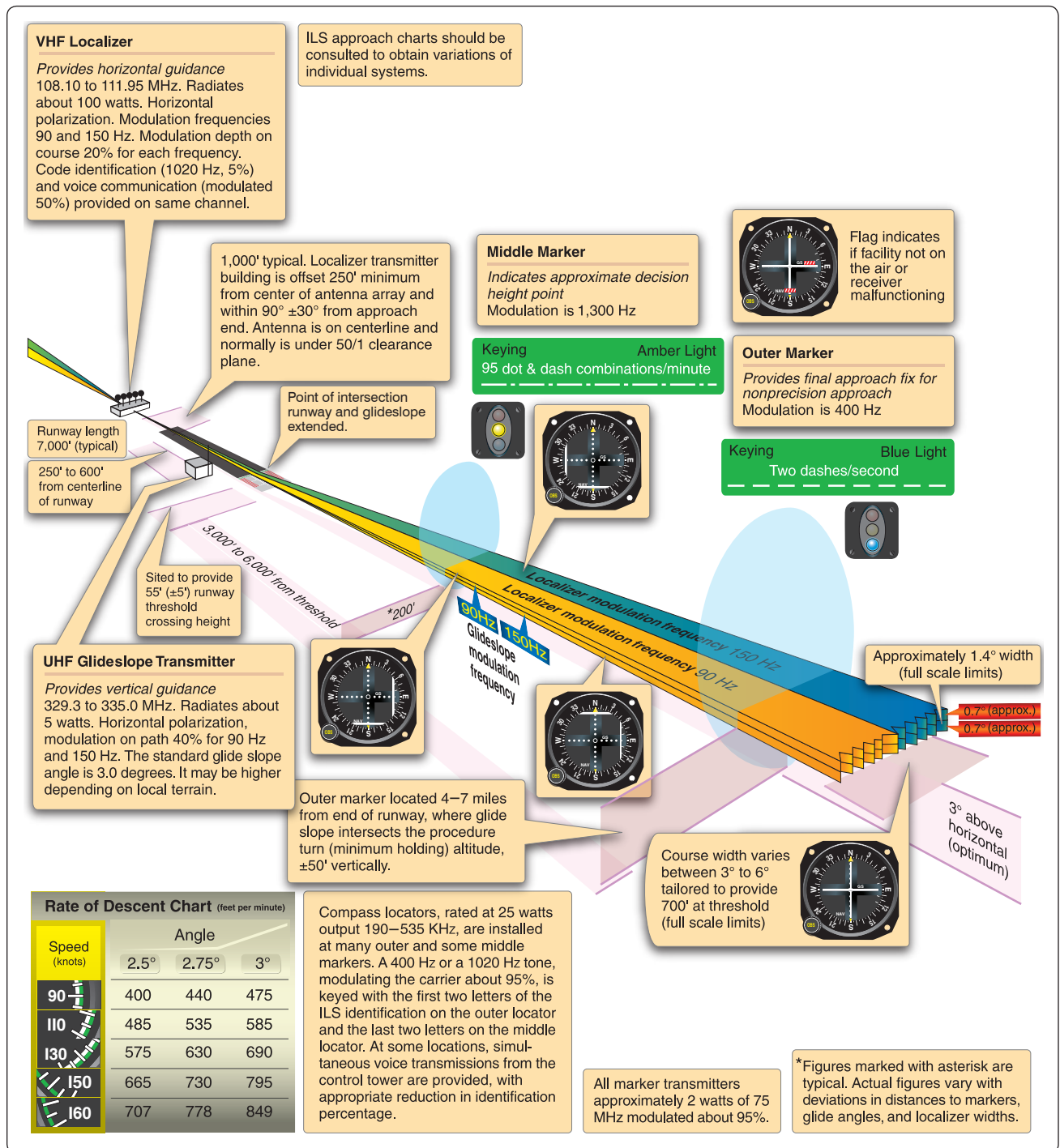


Note: All elevations shown are with respect to the station's site elevation (AGL). Coverage is not available in a cone of airspace directly above the facility.

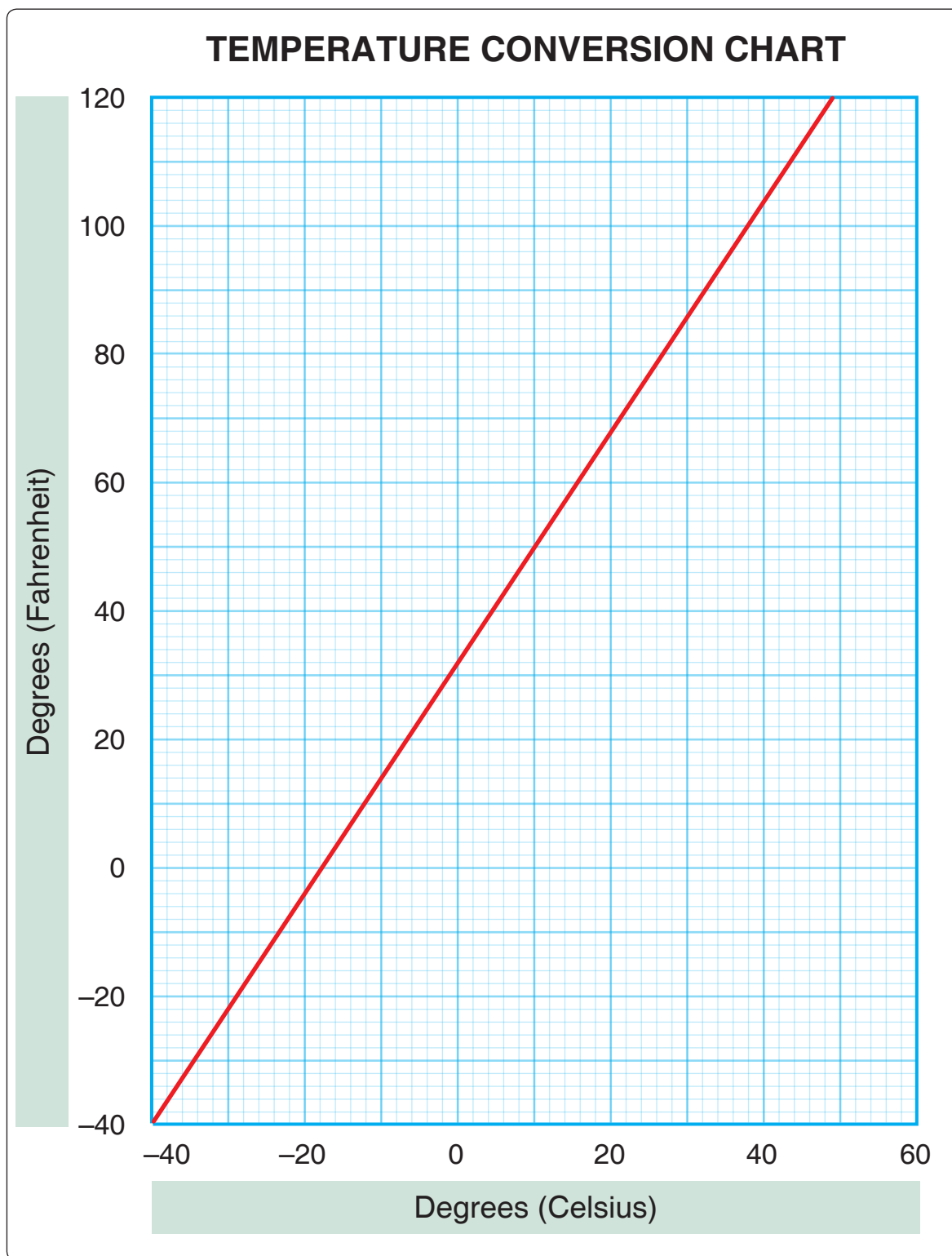
Standard Terminal Service Volume



LEGEND 37.—Air Navigation Radio Aids.



LEGEND 38.—ILS Standard Characteristics and Terminology.



LEGEND 39.—Temperature Conversion Chart.